

FROM SINGLE- TO DOUBLE-PAIR: IMPROVING WI-FI HUMAN ACTIVITY RECOGNITION ON ESP32 DEVICES FOR ELDERLY CARE

*Master Thesis Project - Final Presentation
Big Data Management & Analytics (BDMA)*

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Advisor: Piotr Antonik - CentraleSupélec & ZoeCare CTO*



INTRODUCTION

BACKGROUND

EXPERIMENTAL SETUP

EXPERIMENTS AND EVALUATION

CONCLUSION AND PERSPECTIVES


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CONTEXT & MOTIVATION



Context: 703 M people were aged 65+ in 2019 and about 1.5 B are expected by 2050.



Goal: recognise one resident's daily activities from how the body distorts the Wi-Fi signals using low-cost ESP32 boards: **nothing worn, and no camera.**



Core difficulty: published ESP32 accuracies of 93-97 % are measured in line-of-sight (LOS), in a single room, on pre-segmented clips. Real deployment calls for unseen rooms, non-line-of-sight (NLOS) geometry and a continuous stream of data.



OBJECTIVE, RESEARCH QUESTION & HYPOTHESIS

RESEARCH QUESTION

Can two low-cost ESP32-S3 pairs recognise an elderly's typical daily activities in a room the model has never seen, under non-line-of-sight (NLOS) conditions & how much does the second pair really contribute?

OBJECTIVE

One specific configuration: two ESP32-S3 transmitter-receiver pairs in the same room, with the subject in non-line-of-sight of both links.

To our knowledge, this **geometry had not been studied** before.

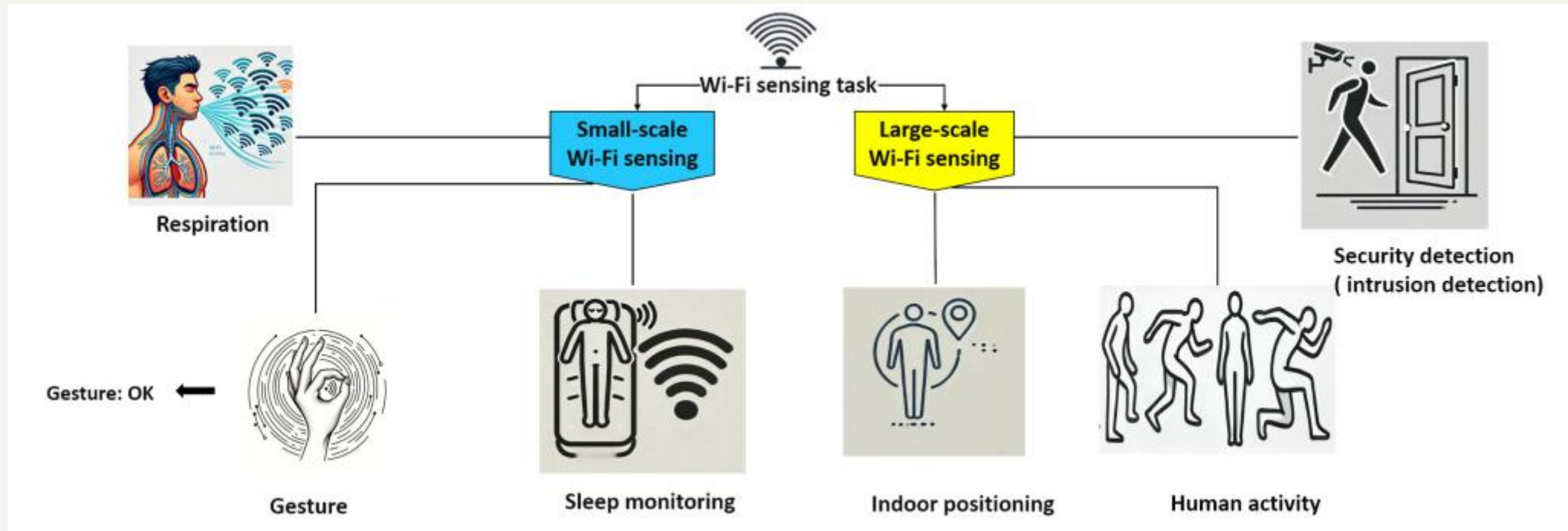
HYPOTHESIS

A Wi-Fi link responds strongly to motion that crosses it and weakly to motion that runs along it.

So two orthogonal links should cover each other's blind spots.

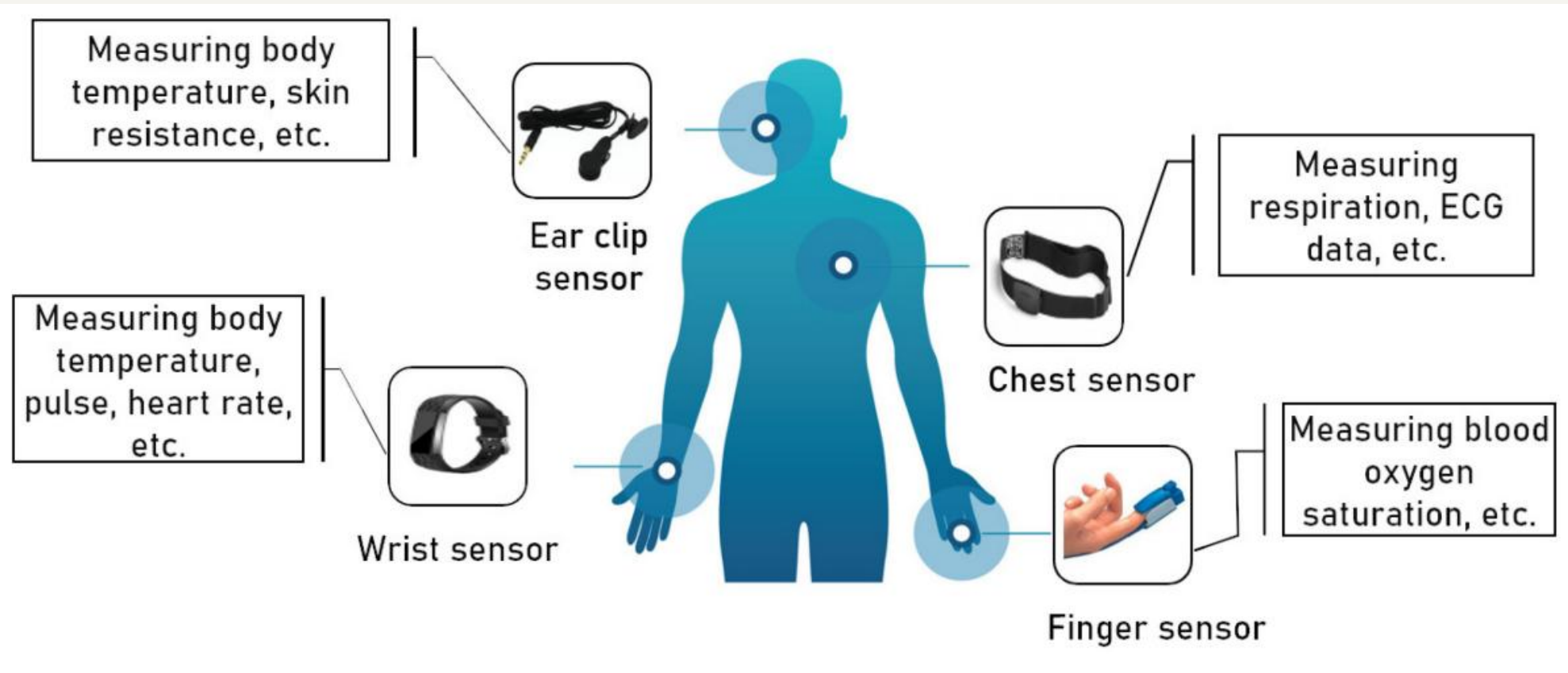


Wi-Fi sensing taxonomy



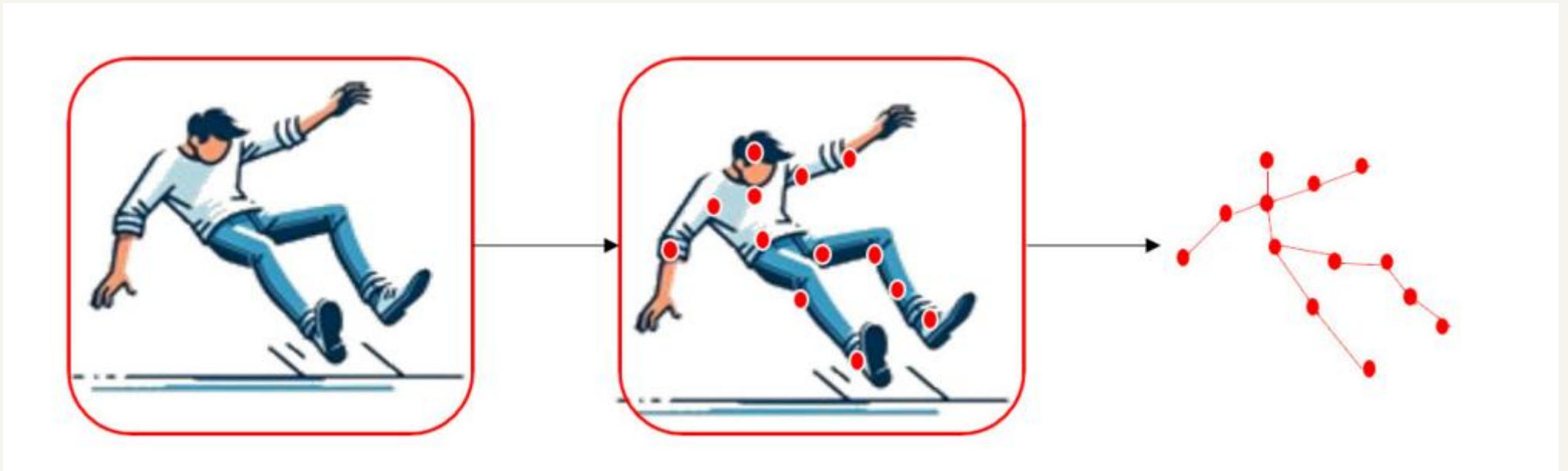
EXISTING APPROACHES HAR

HAR modalities: sensor / wearable (accurate but intrusive - battery, drift)

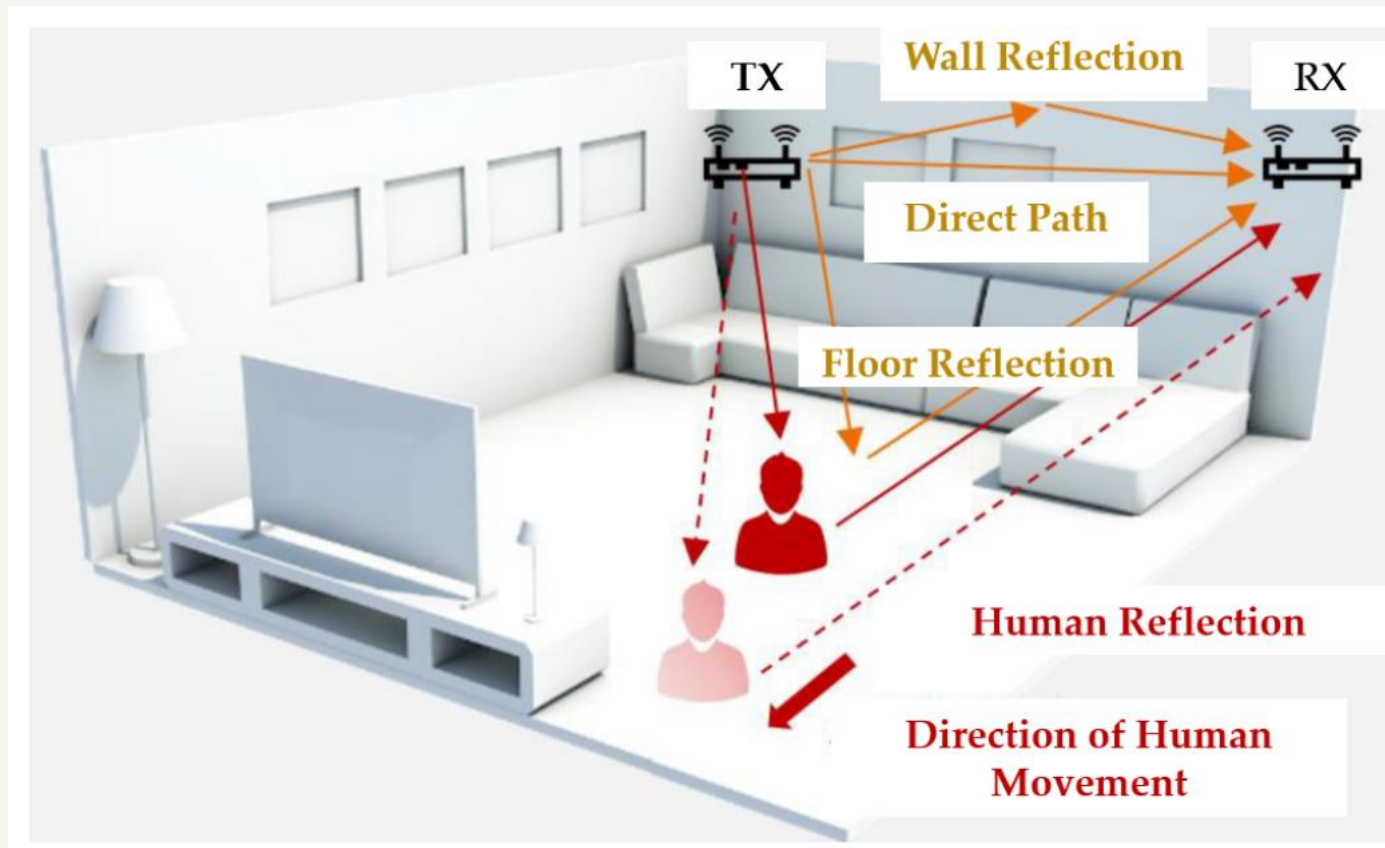


EXISTING APPROACHES HAR

HAR modalities: camera / vision (rich but privacy- and lighting-sensitive)



HAR modalities: Wi-Fi - CSI (device-free, privacy-friendly, reuses existing infrastructure).



Reported ESP32 result	Accuracy	Conditions	What the number hides
LSTM on Raspberry Pi	93 %	daily activities + falls, LOS	single room, pre-segmented clips
Quantised GRU	93.5 %	7 classes, LOS	same-room evaluation only
Long-range presence detection	≈ 97 %	2 classes, line-of-sight	binary task, LOS only
ESP-Fi HAR benchmark (reproduced here)	76.1 %	4 rooms, leave-one-room-out	10 s windows, subject-mixed splits

→ *Headline ESP32 scores come from line-of-sight, single-room, pre-segmented clips. The open gap is cross-environment NLOS recognition on short windows in a continuous stream & exploiting several diversely placed links is itself named an open challenge.*

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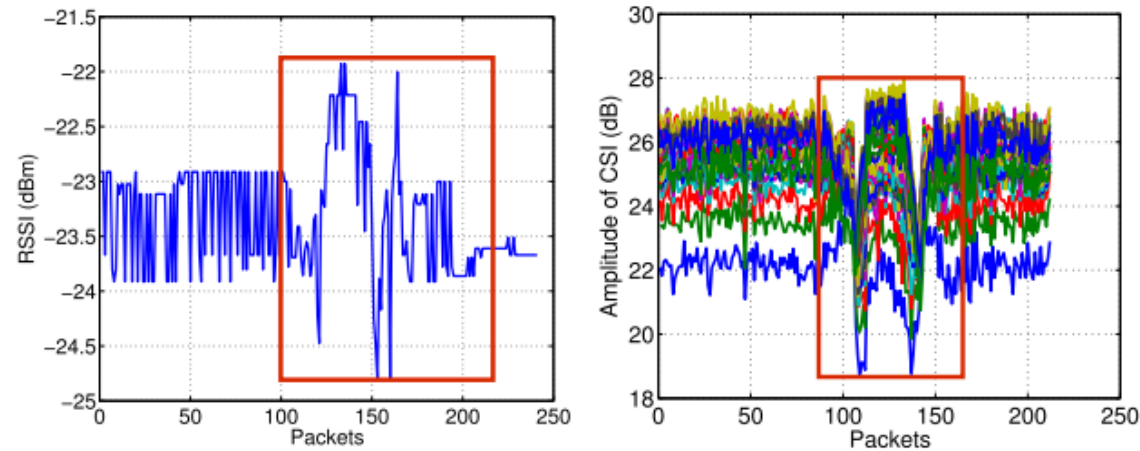

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Wi-Fi CSI

How the channel transforms the Wi-Fi signal across OFDM subcarriers.

- Amplitude + phase per subcarrier (RSSI gives only a coarse average)
- Body motion perturbs multipath reflections → measurable change
- Fine-grained: from walking down to breathing depending on device and configuration



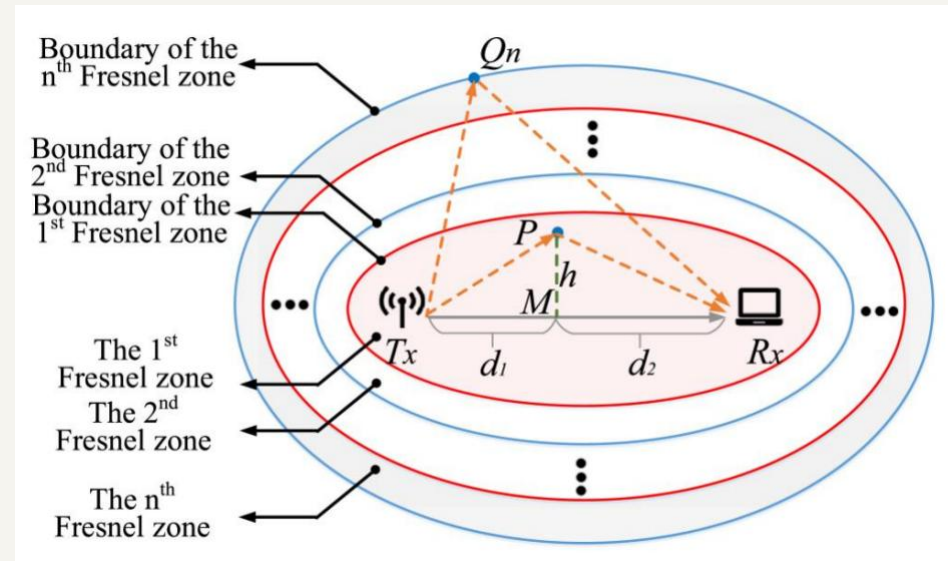
(a) RSSI

(b) CSI

Fresnel Zones & Orientation

A single link is not uniformly sensitive - its blind spots are geometric, not algorithmic.

- **Fresnel zones:** ellipsoids with TX and RX at the two foci. For a 4 m link at 2.4 GHz ($\lambda \approx 0.125$ m) its radius at the midpoint is only ≈ 0.35 m.
- **Orientation matters as much as position**
- **Two pillars for two pairs: orthogonal links view the area from different angles, so motion poorly observed by one link tends to be well observed by the other.**



Fresnel-zone geometry: TX and RX sit at the foci of a family of confocal ellipses. The first zone (innermost) carries over 70% of the transmitted energy.

ESP32-S3

Low-cost Wi-Fi microcontroller that exposes raw CSI.

- Cheap, low-power and widely available
- Per-packet CSI on commodity hardware (52 subcarriers)
- Tight memory / compute budgets → lightweight models only

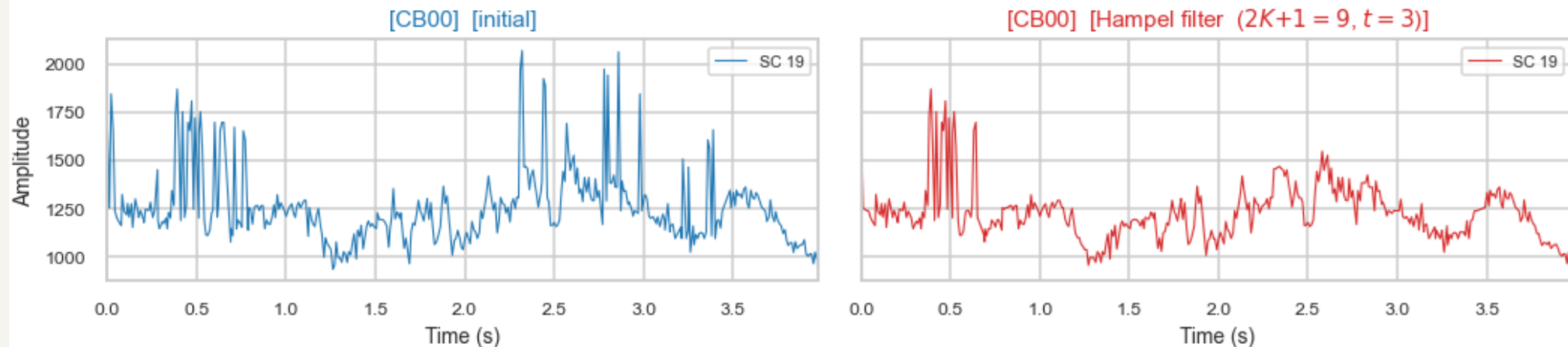


Hampel filter

Why: impulsive noise - isolated spikes and dips caused by the single antenna, power-amplifier uncertainty, interference and rate adaptation.

- **The Hampel filter acts only where the data is anomalous**
- **Result:** hardware spikes go, genuine fast dynamics stay.

Hampel Filter example for $K=4$



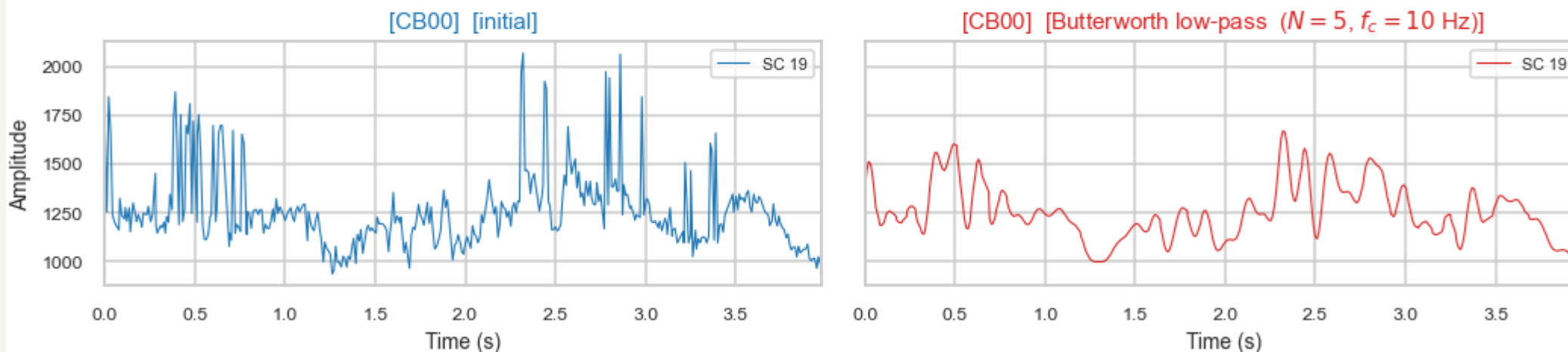
DENOISING FILTERS - BUTTERWORTH BANDPASS

Butterworth bandpass

Why: with the spikes removed, low-level Gaussian noise remains - and human activity perturbs the channel only at comparatively low frequencies.

- **Keep the motion band, discard the rest.**
- **The order N sets how steeply everything past the cut-off is attenuated.**
- **One of the most important steps in our final pipeline**

Butterworth low-pass filter example



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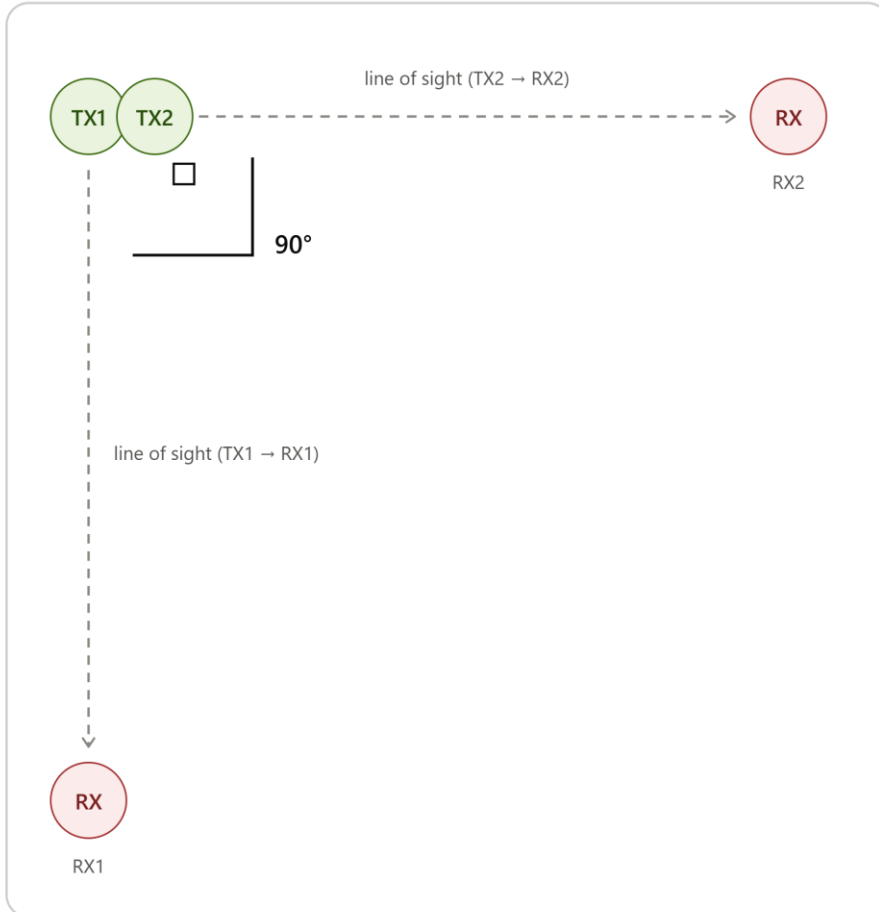
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Floorplan



Physical configuration

RX Naming	TX Height	RX Height	RX-TX distance
A) RX1: AC84	140 cm	95 cm	3.8 m
B) RX2: CB00	140 cm	55 cm	3.5 m



Acquisition - four ESP32-S3 boards, two orthogonal links:

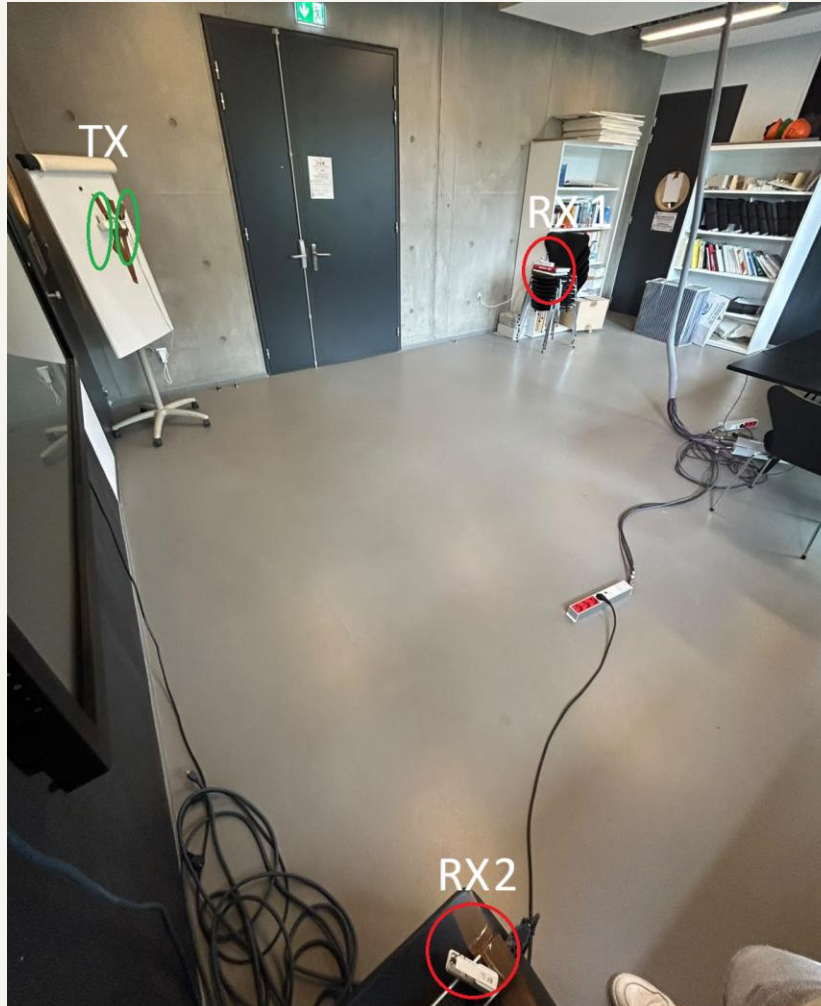
- Two transmitters co-located
- 52 subcarriers per link at ≈ 100 Hz capturing amplitude only.
- Synchronised start, so both links share one wall-clock timebase



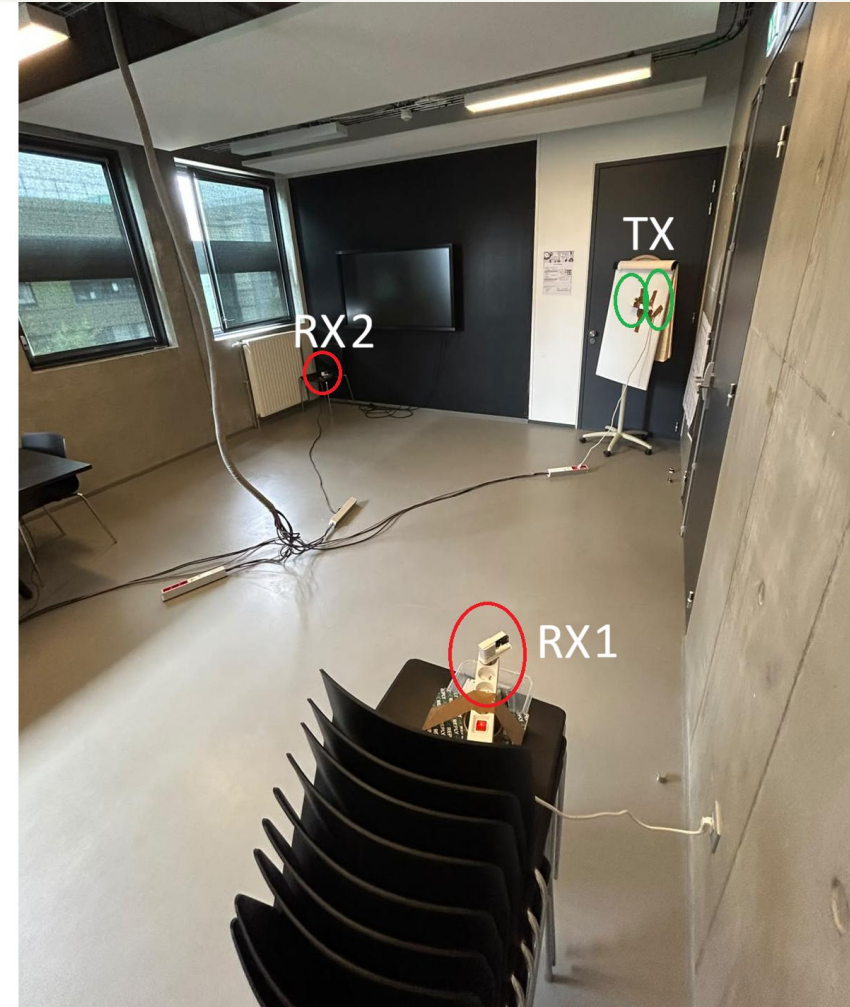
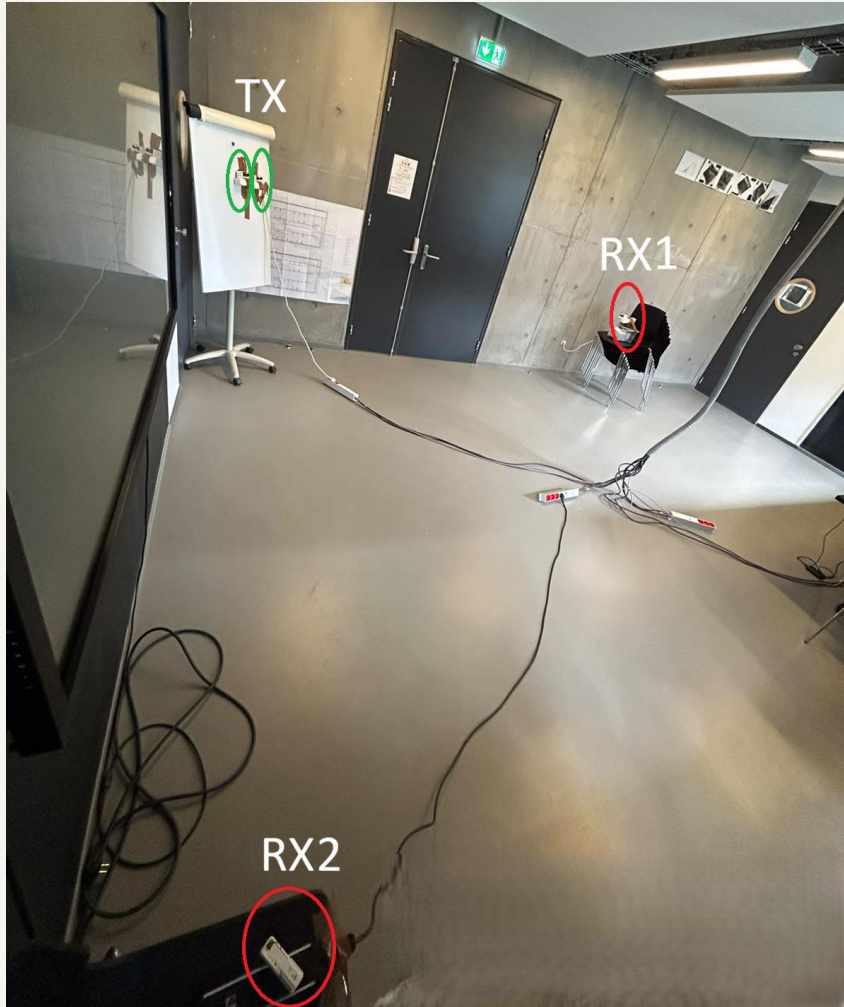
Protocol - six classes, recorded as movements (transitions) rather than held postures

- walk, prepare_meal, sit, sit_up (stand up) , lying, lying_up (getting up)
- Cued clips: each recording starts and ends on a beep with exactly one activity inside
- **Three rooms** (mc208, sb207, mf207) over several days, ≈ 420 labelled windows per room
- Subject in non-line-of-sight of both links (a single subject)

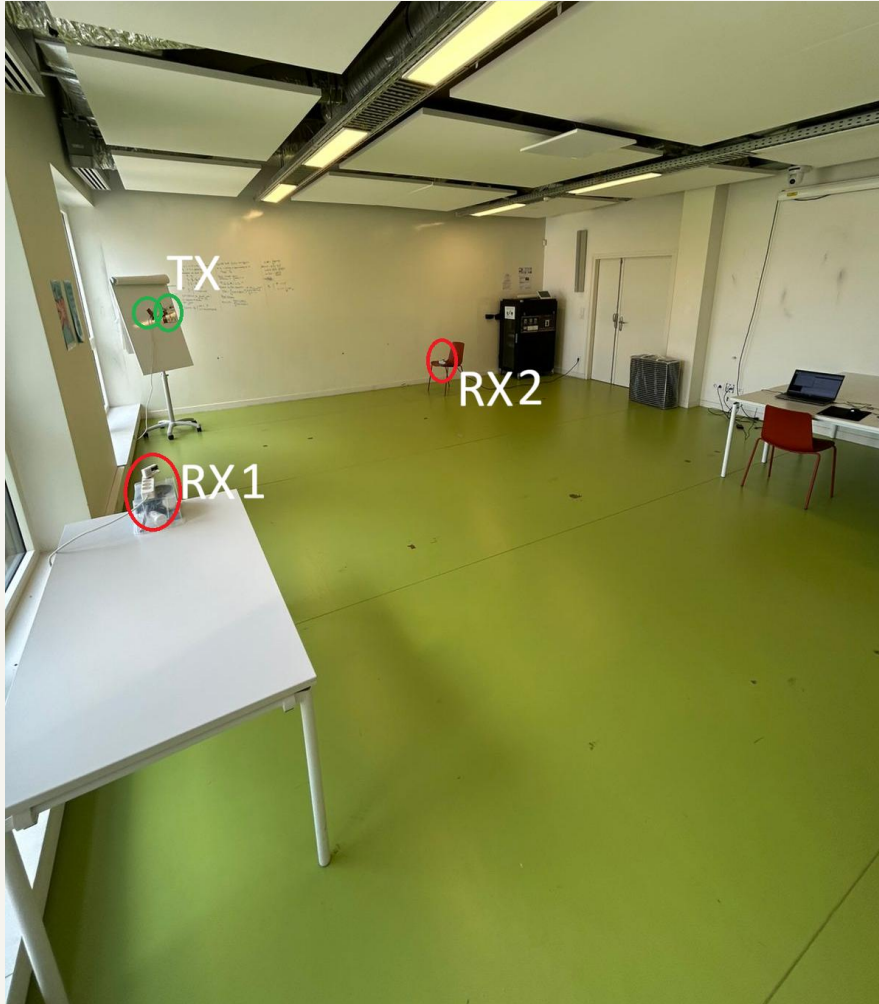
EXPERIMENTAL SETUP – ROOM MC208

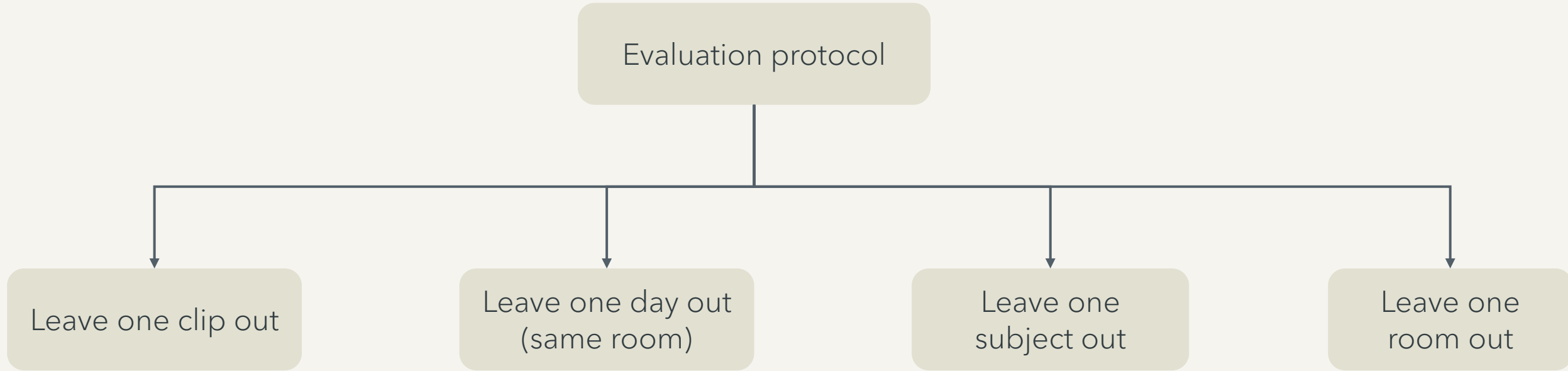


EXPERIMENTAL SETUP - ROOM MF207

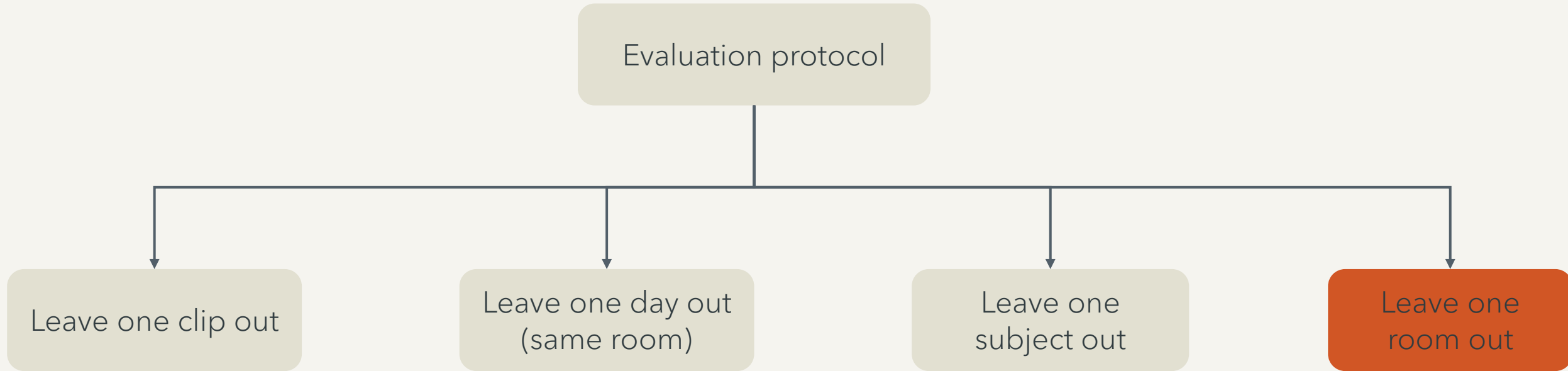


EXPERIMENTAL SETUP - ROOM SB207





The split always isolates the variable under test



The most difficult split is leave-one-room-out (cross-environment)



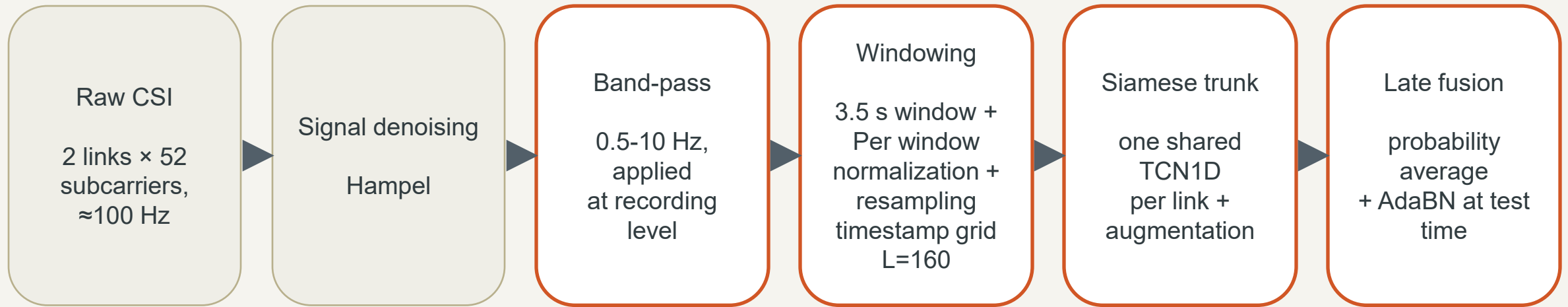
Protocol:

- Leave-one-room-out: train on **two** rooms, test on the **third**, never seen
- Score = mean macro-F1 over the three folds
- Beyond cross-validation: live sessions in a fourth unseen room, and long continuous recordings with no segment boundaries



Granularities:

- Six classes: walk, prepare_meal, sit, lying, sit_up, lying_up
- Four class where (sit+lying → going down, sit_up+lying_up → getting up)
- **Extension:** addition of a no_activity class



Orange outline = stage introduced or changed in this thesis.

My contributions in this pipeline

- **Architecture:** each link is encoded by the same siamese trunk and fused late, instead of stacking both links into one 104-channel input
- **Windowing:** 3.5 s windows resampled onto a wall-clock timestamp grid and downsampled into L=160 steps
- **Augmentation:** subcarrier and time masking and same-class mixup
- **Adaptive BN:** computer vision idea to achieve unlabelled test-room calibration by reestimating BN running statistics

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EXPERIMENT I – BENCHMARK REPRODUCTION: SETUP

GOAL

Establish a reference point: *how far does a single ESP32 pair generalise to an unseen room, using published data and published models?*

CONDITIONS

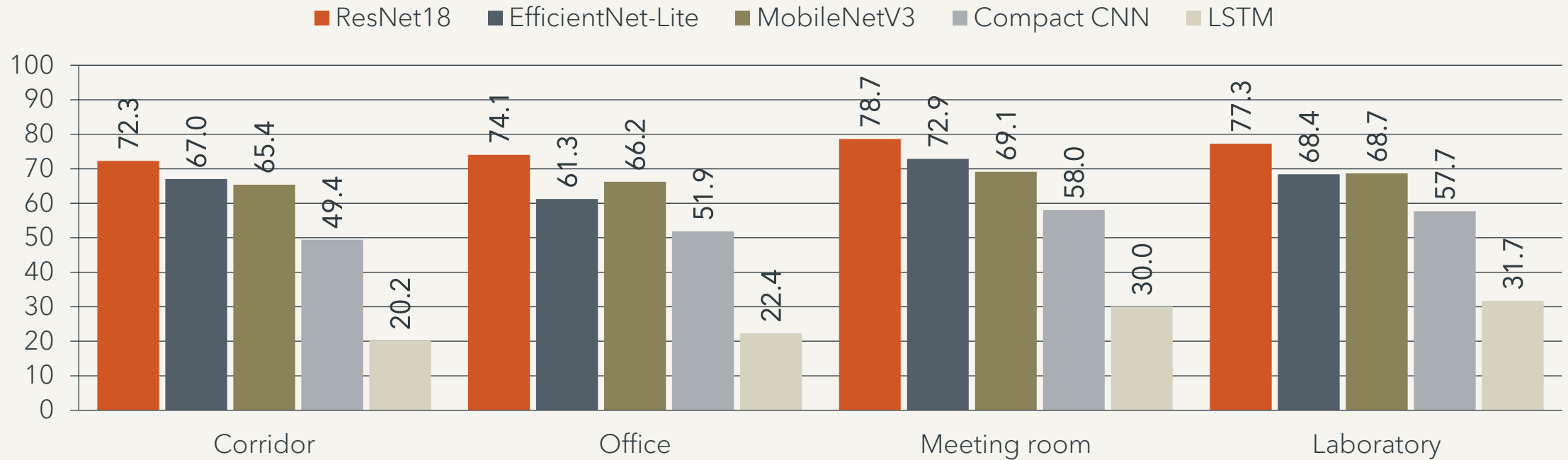
- Public ESP-Fi HAR dataset - 8 subjects, 4 rooms, 7 classes
- Mostly line-of-sight (based on floorplan), single TX-RX pair, 10s windows
- Published models, retrained end to end by us

WHAT WE DID

- Re-ran the published ESP-Fi HAR benchmark on its public dataset with its own models, retrained from scratch
- **The aim was to fix the reference point: how well device-free HAR currently performs on ESP32-class hardware with a single TX-RX pair.**

BENCHMARK REPRODUCTION - CROSS-ROOM GENERALIZATION

Leave-one-environment-out macro-F1 (%) on the held-out room - chance level 14.3%



- **ResNet18 leads throughout** : 75.6 mean macro-F1 (76.1% accuracy), and EfficientNet-Lite comes second.
- **It depends on the backbone, not the protocol**: convolutional models transfer, while the recurrent ones fall to 20-32 macro-F1, barely above the 14.3% seven-class chance level.



PRELIMINARY EXPERIMENTS - CROSS-DAY, NLOS

GOAL

*Which activities can **two co-deployed pairs** separate at all, when the same room is recorded on different days and both links are in NLOS?*

CONDITIONS

- One room, four days, leave-one-day-out
- Two activities: prepare meal and walk
- Four static activity states: standing, sitting, lying, empty room
- All activities are performed in non-line-of-sight

RESULTS

- Both ML families plateaued: 0.558 macro-F1 (classical features) and 0.451 (best CNN)
- Merging the four static states into one no_action label lifted both families to 0.911 / 0.956 F1 under the same protocol
- **Negative result that reshaped the thesis: a motionless body leaves no motion-band signature, so every class must carry movement. The main experiment records posture transitions instead of held postures.**



PRELIMINARY EXPERIMENTS - FINE-GRAINED KITCHEN HAR

GOAL

How fine can the activities get - and does a score measured on controlled clips still hold on continuous, naturally performed routines?

CONDITIONS

- Double pair TX-RX
- Mock kitchen, LOS, same room and day
- Four fine-grained counter actions + walk, stand
- Clips versus continuous real routines

RESULTS

- On controlled single-activity clips (leave-one-out, 198 clips): 0.909 accuracy and 0.909 macro-F1
- The same deployed pipeline on six continuous coffee- and tea-making routines (350 windows): 0.589 accuracy, 0.445 macro-F1
- **Same pipeline, same room, same day - only the evaluation regime changed. Segmented clip scores measure separability, not deployability.**



MAIN EXPERIMENT - CROSS-ENVIRONMENT GENERALIZATION

INITIAL RESULTS

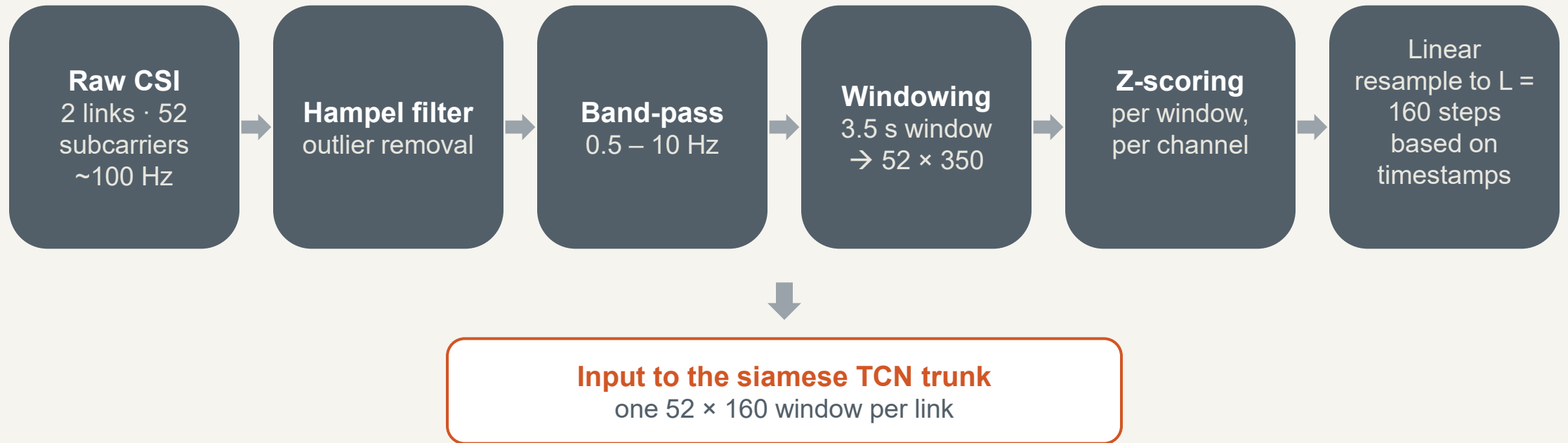
- A broad grid search over both model families plateaued at 0.579 macro-F1 → the limit was structural, not hyper-parameters
- A plethora of experiments was implemented to break this plateau and it lead to the final pipeline

GOAL

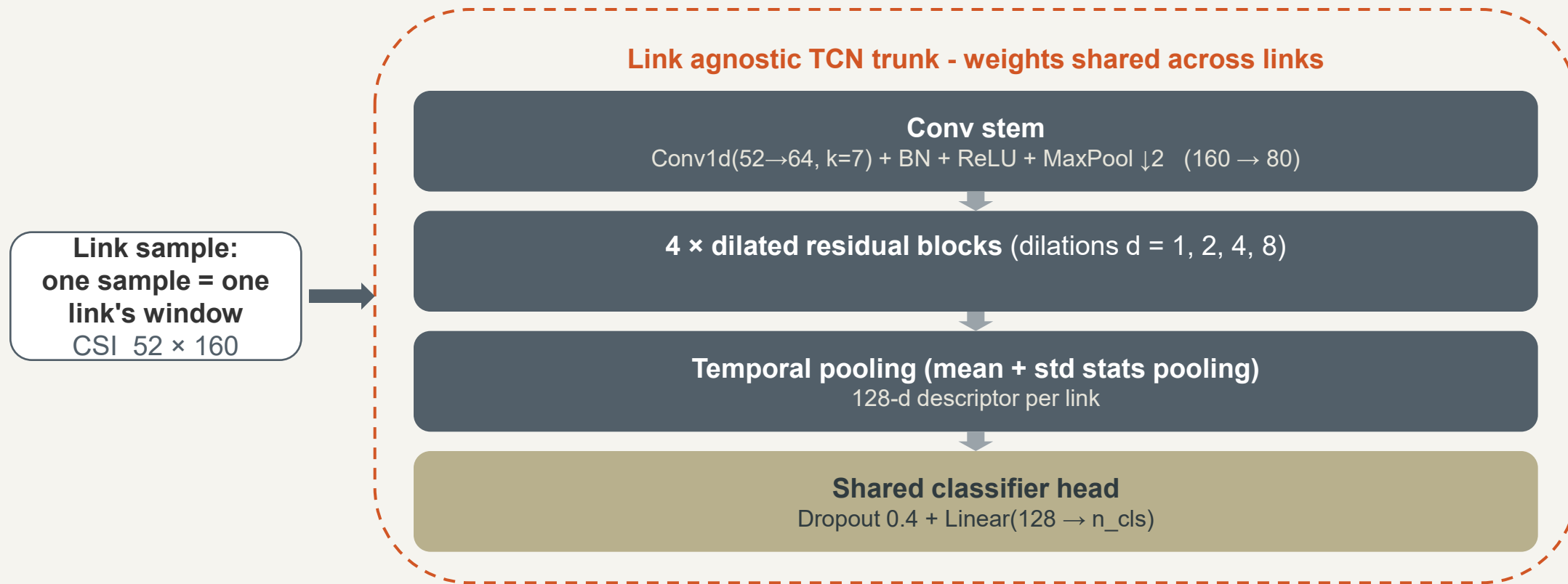
Can a model trained in two rooms work in a third it has never seen and what makes that possible, and is the second pair worth its cost?

CONDITIONS

- Double pair TX-RX
- Three rooms, leave-one-room-out, ≈420 windows each
- Six transition classes: walk, prepare meal, sit, lying, sit_up, lying_up



FINAL MODEL – ARCHITECTURE - TRAINING



TRAINING RECIPE

Optimisation AdamW, lr $1e-3$, weight decay $1e-3 \cdot 30$ epochs · batch 32

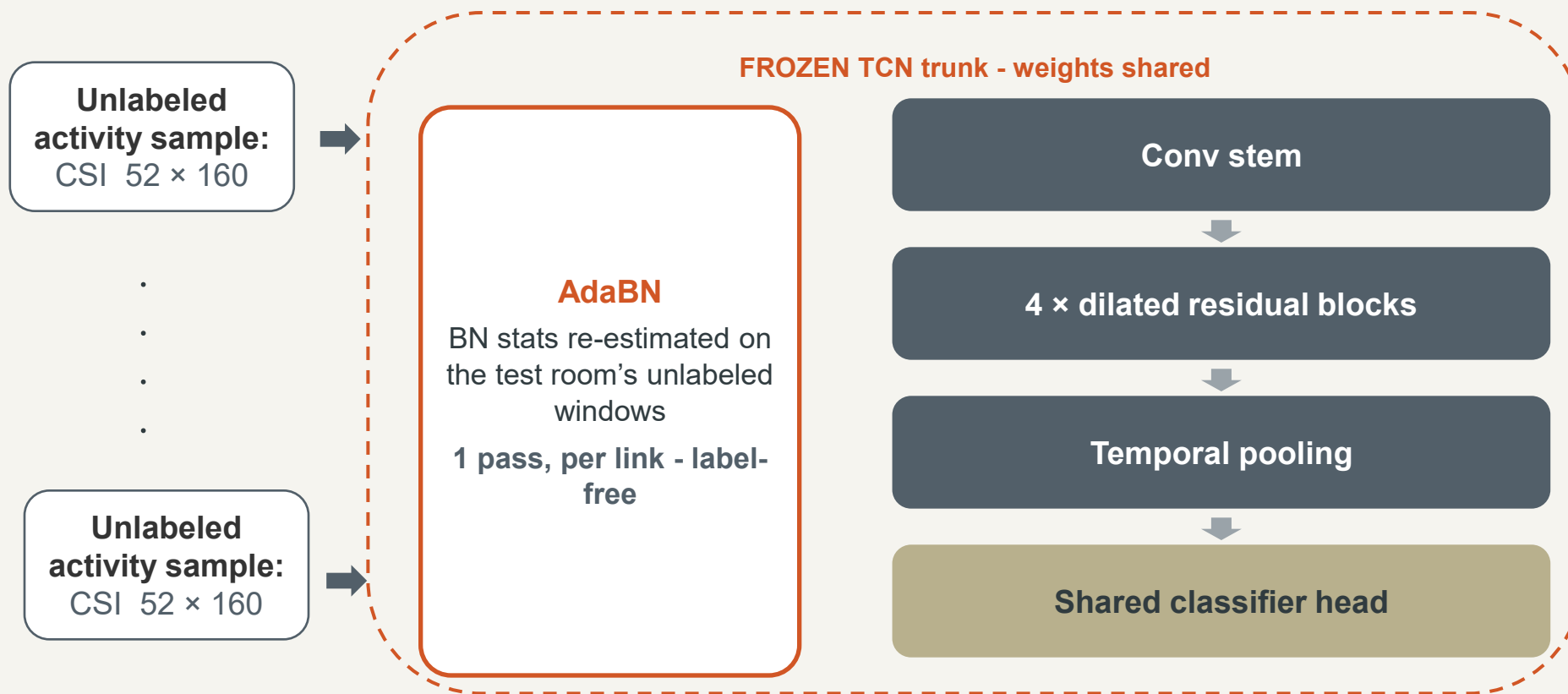
Schedule / loss cosine annealing; cross-entropy with label smoothing 0.05

Masking subcarrier + time masking, $p = 0.7$ (3–8 subcarriers / spans of 8–20 steps)

Same-class mixup $p = 0.5$, $\lambda \in [0.6, 0.95]$ - partner may come from the other link: a deliberate cross-geometry regulariser

3-seed ensemble

FINAL MODEL – UNLABELLED TEST ROOM ADAPTATION



AdaBN happens per link so two links are no longer the same network: same weights but different BN statistics



FINAL MODEL – ARCHITECTURE INFERENCE

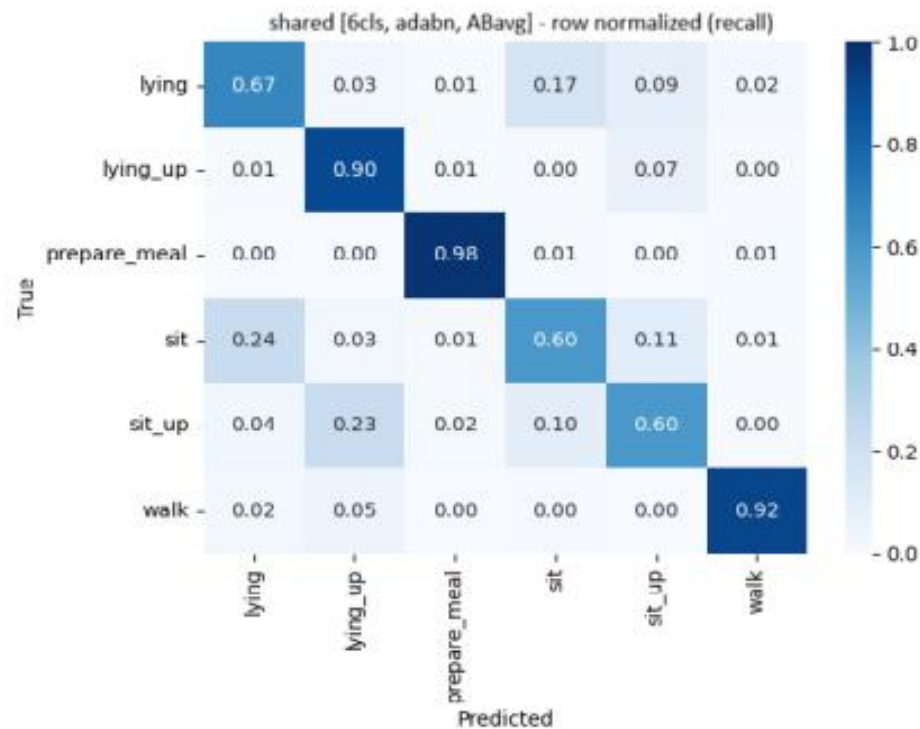


MAIN EXPERIMENT – CONFUSION MATRICES DOUBLE PAIR

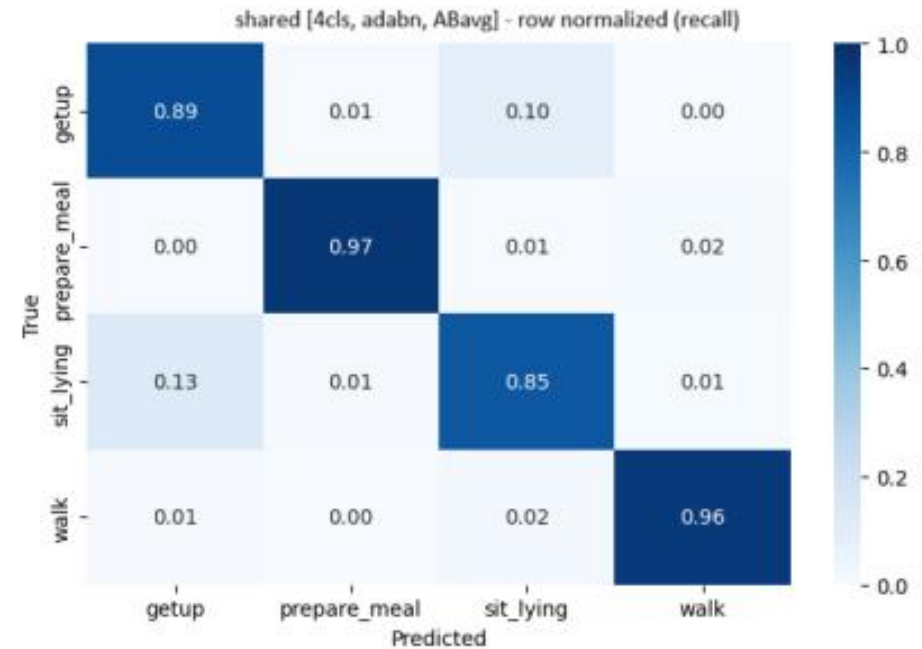
→ The confusion between chair and floor is clear even for the double pair

→ 0.766 on 6-class and 0.918 macro-fi on 4-classes

6-CLASS



4-CLASS





IS THE SECOND PAIR WORTH ITS COST?

Training	Test-time input	6-class F1	4-class F1
Shared trunk (both links)	Both links, probability average	0.766	0.918
Shared trunk (both links)	Link B only	0.724	0.888
Single link B	Link B only	0.686	0.867
Shared trunk (both links)	Link A only	0.677	0.869
Single link A	Link A only	0.607	0.827

- Averaging the two links beats the best single-link deployment by 4.2 points (six classes) and 3.0 points (four classes).
- A link trained inside the pooled shared trunk beats the same link trained alone by 3.8 (link B) and 7.0 points (link A)
- **The practical recipe is collect with two pairs, deploy with one.**

EXTENDING THE CLASS SET WITH A NO_ACTION CLASS

Training	Test-time input	7-class F1	5-class F1
Shared trunk (both links)	Both links, probability average	0.777	0.917
Shared trunk (both links)	Link B only	0.734	0.881
Single link B	Link B only	0.711	0.862
Shared trunk (both links)	Link A only	0.692	0.848
Single link A	Link A only	0.627	0.830

- **The idle class is nearly free:** 0.777 macro-F1 on seven classes and 0.917 on five, against 0.766 and 0.918 without it → the other classes pay almost nothing for the extra label.
- **Every conclusion survives the extra class:** late fusion of both links still leads, the pooled shared trunk still lifts a single link, and link B still beats link A. Collect with two pairs, deploy with one.



REAL-TIME DEPLOYMENT IN UNSEEN ROOMS

Live session (room never seen in training)	Input at test time	7-class macro-F1 / acc	5-class macro-F1 / acc
Session 1: fourth unseen room	single pair	0.779 / 0.783	0.899 / 0.883
<i>Offline leave-one-room-out estimate</i>	<i>single pair</i>	<i>0.734 / 0.740</i>	<i>0.881 / 0.881</i>
Session 2: fifth unseen room	double pair	0.801 / 0.800	0.948 / 0.943
<i>Offline leave-one-room-out estimate</i>	<i>double pair</i>	<i>0.777 / 0.785</i>	<i>0.917 / 0.918</i>

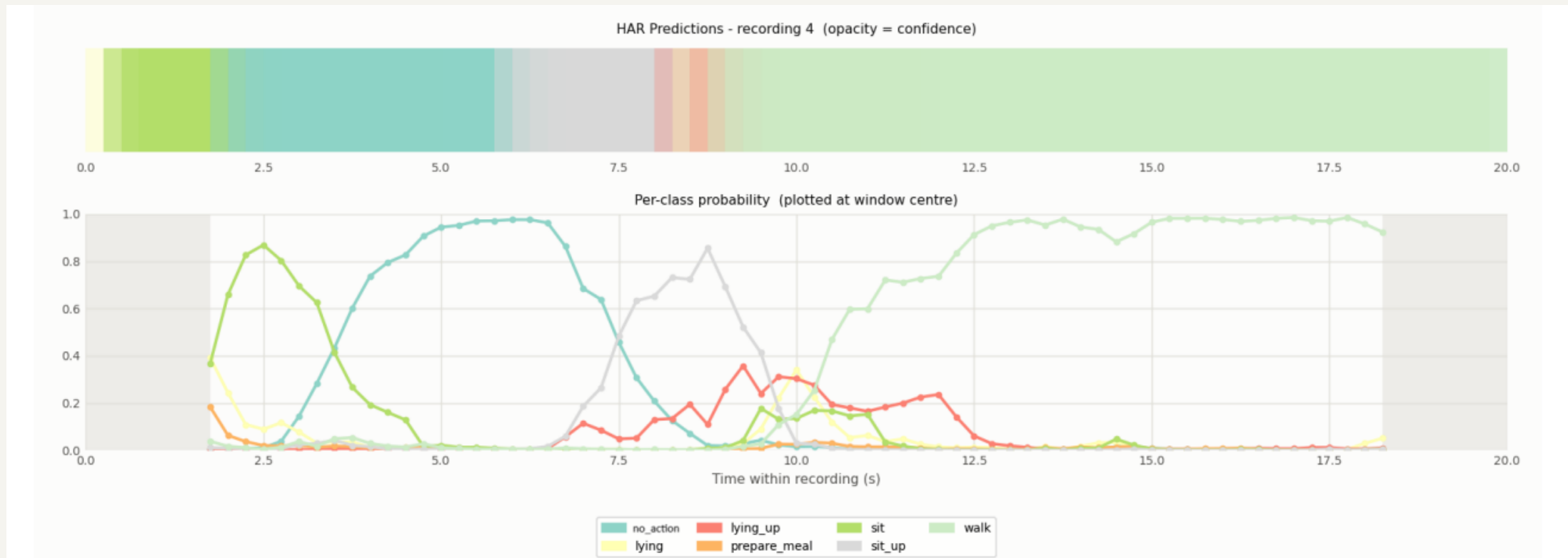
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- **Calibration is cheap:** AdaBN only re-estimates batch-norm statistics from unlabelled windows, so adapting to a new room means one cued window per class
- **The live scores land slightly above the offline estimate,** this is a genuine held-out evaluation of the pipeline; the small gain is consistent with the larger training set used (all three rooms)
- **The two live sessions differ in room, day and calibration,** so they show the pipeline works in each setting

REAL TIME CONTINUOUS DEPLOYMENT EXAMPLE

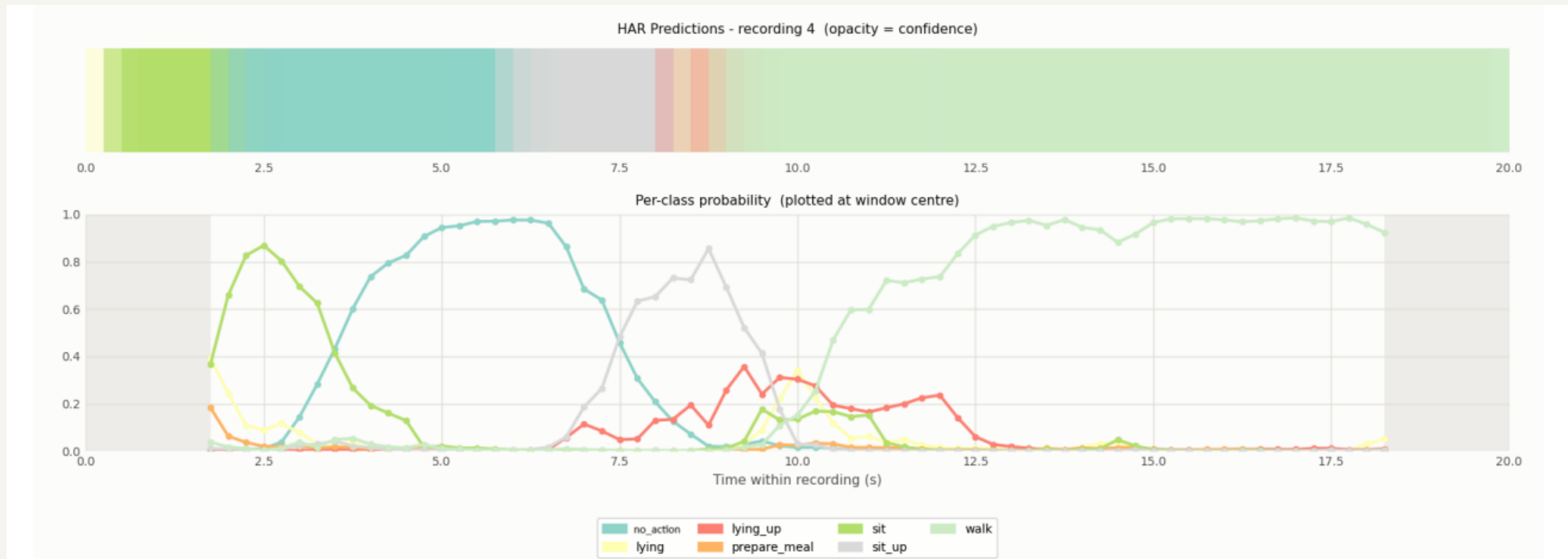
→ GT: lying → no_action → lying_up → walk

→ Lying is slightly confused with sitting in the beginning and lying_up is confused with sit_up after the idle state



REAL TIME CONTINUOUS DEPLOYMENT EXAMPLE

- **On continuous recordings the model did not collapse:** walk, prepare_meal, no_action and most posture transitions were still recognised, and the errors concentrate in the transitions between two activities.



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CONCLUSION – ANSWERING THE RESEARCH QUESTION

THE ANSWER

Cross-environment recognition works in NLOS with two low-cost or one ESP32-S3 pair.

0.724 / 0.888

six-class / four-class macro-F1

Leave-one-room-out, in rooms never seen using **single pair** TX-RX on inference.

0.766 / 0.918

six-class / four-class macro-F1

Leave-one-room-out, using the **double pair**

- **The second pair pays off mainly at collection time.** Recipe: collect with two pairs, train with both pairs data, deploy with one.
- **Link placement matters as much as how many links there are:** Link B consistently outperformed link A
- **Deployment checks: An idle no_action was successfully added and the pipeline sustained performance in realtime and did not collapse in real time continuous recordings**

LIMITATIONS

- **A single subject**, the results are cross-room but same-subject
- **Three rooms**, a few hundred labelled windows each.
- **Configurations ranked on the held-out folds** and therefore slightly optimistic.
- **Training data is cued clips**
- **The chair-versus-floor confusion**: that the four class granularity avoids

PERSPECTIVES

1. **Investigate separation of sitting from lying**
2. **Collecting transition data for continuous inference**
3. **More rooms, and more than one subject**
4. **Exploring TX-RX placements systematically**: height and distance
5. **Towards deployment**: on-device inference, automatic AdaBN calibration and server upload.

Questions & discussion

Stefanos Kypritidis - Big Data Management & Analytics (BDMA)

Advisor: Piotr Antonik - CentraleSupélec & ZoeCare CTO



- Evolution of Wifi sensing
- HAR sensing modalities
- Device info
- Median filter example
- Wavelet filter example

Methodology:

- Methodology part 1
- Methodology part 2
- Activity classes detailed

Preliminary:

- Preliminary experiment 1 results
- Preliminary experiment 2 results

Main:

- Main experiment – grid & hand crafted

- Preprocessing pipeline
- Architecture training
- Adaptive BN
- Inference plot
- Ablation analysis
- Temporal strategy grid
- Evolution of f1 per experiment
- Confusion matrix double pair
- Classification report double pair
- Confusion matrix single pair B
- Per room and approach results
- No action class extension
- Live session results
- Live session confusion matrix
- Continuous sessions examples

THE EVOLUTION OF WI-FI SENSING

PC-CLASS HARDWARE - INTEL 5300 NIC

Device-free WiFi sensing
detect people without
wearables or cameras

2007

RSSI → CSI
amplitude + phase
across OFDM
subcarriers

2013

WiAR dataset
one of the first open
Wi-Fi HAR datasets;
CNN/LSTM >90%

2019

2011
Intel 5300 CSI tool
commodity 802.11n
CSI, fine-grained
sensing

2015
CARM
physics-based activity
models; 96% on Intel
5300

EMBEDDED EDGE - ESP32

ESP32-S3 HAR
through-wall 97% /
88%; one of the first
ESP32 datasets

2023-24

2020
ESP32 exposes CSI
sensing moves via low-
cost IoT edge

2025-26
Edge HAR & the gap
cheaper, lower-power -
but cross-environment
still drops → our double
pair

RELATED WORK – HAR SENSING MODALITIES

Modality	Accuracy	Privacy	Burden on the user	Why it fails / fits for elderly care
Wearable / IMU	High	Good	Must be worn and charged	Forgotten; usually needs a companion smartphone app
Camera / vision	High	Poor	None	Needs light, cannot see through walls, unacceptable in a bathroom
Ambient motion (PIR)	Low	Good	None	Reports that something moved, never what was done
Wi-Fi CSI (our choice)	Moderate–high	Good	None	Device-free, passes through walls, reuses existing infrastructure

→ *Wi-Fi CSI is the only modality that is at once device-free, privacy-preserving and able to sense through walls - the price is that it is the least robust to a change of room.*



ESP32-S3

Low-cost Wi-Fi microcontroller that exposes raw CSI.

- Cheap, low-power and widely available
- Per-packet CSI on commodity hardware
- Tight memory / compute budgets → lightweight models only

Component	Specification
Product	XIAO ESP32-S3
Processor	ESP32-S3R8 (Xtensa [®] LX7 dual-core, 32-bit up to 240 MHz)
External Memory	8 MB PSRAM, 8 MB Flash
Wireless	2.4 GHz Wi-Fi (802.11 b/g/n), Bluetooth LE 5.0
Dimensions	21 × 17.8 mm
Operating Voltage	3.8 V (battery)/ 5 V (Type-C)
Interfaces	1× UART, 1× IIC, 1× SPI, 11× GPIO, 9× ADC

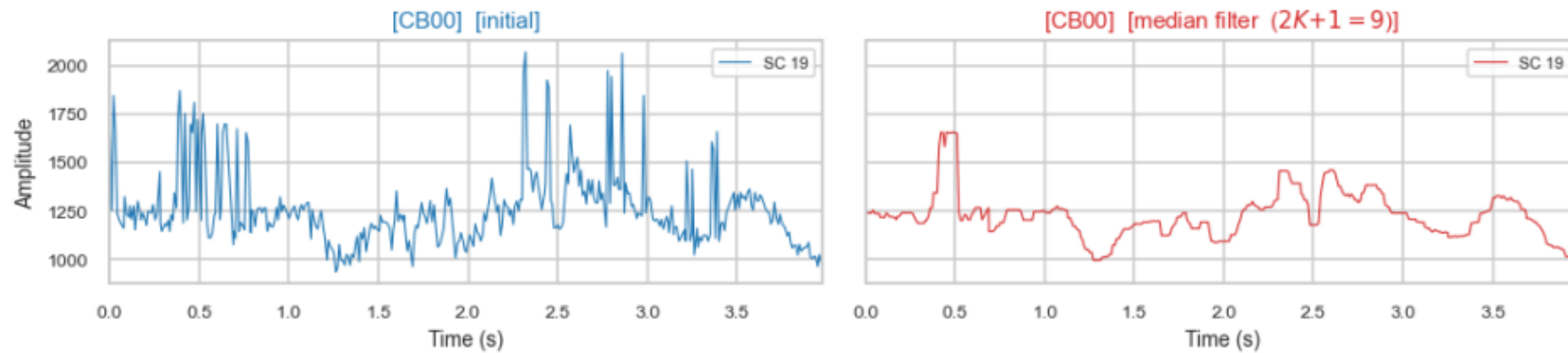
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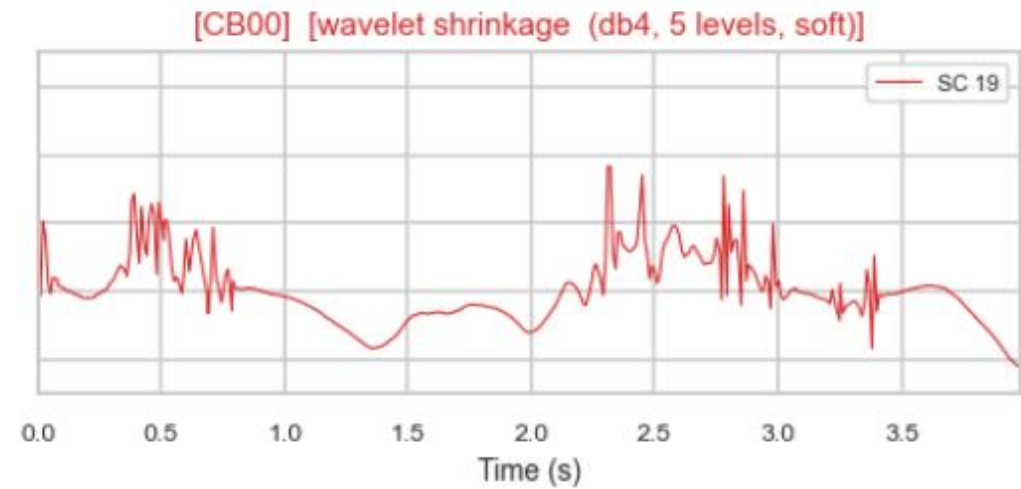
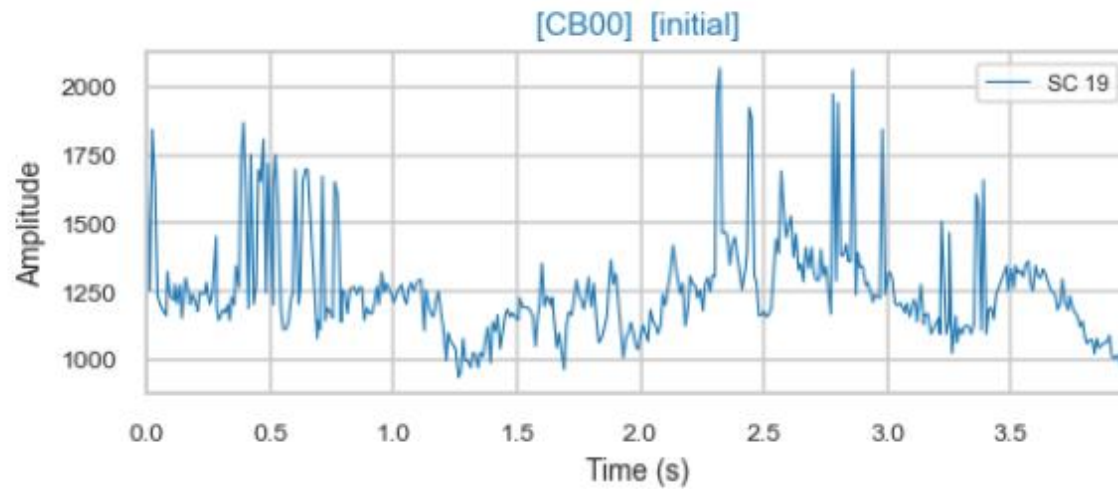
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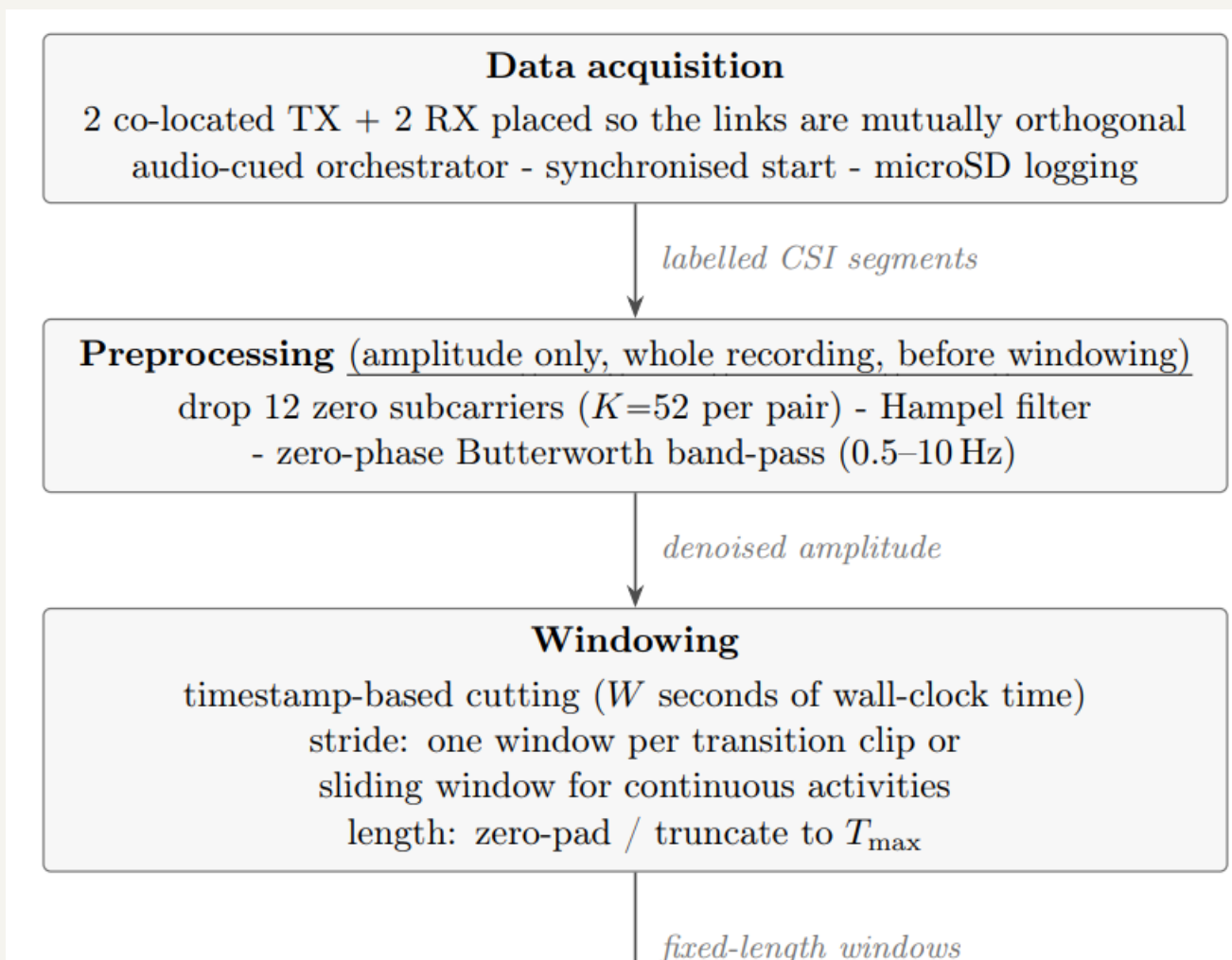


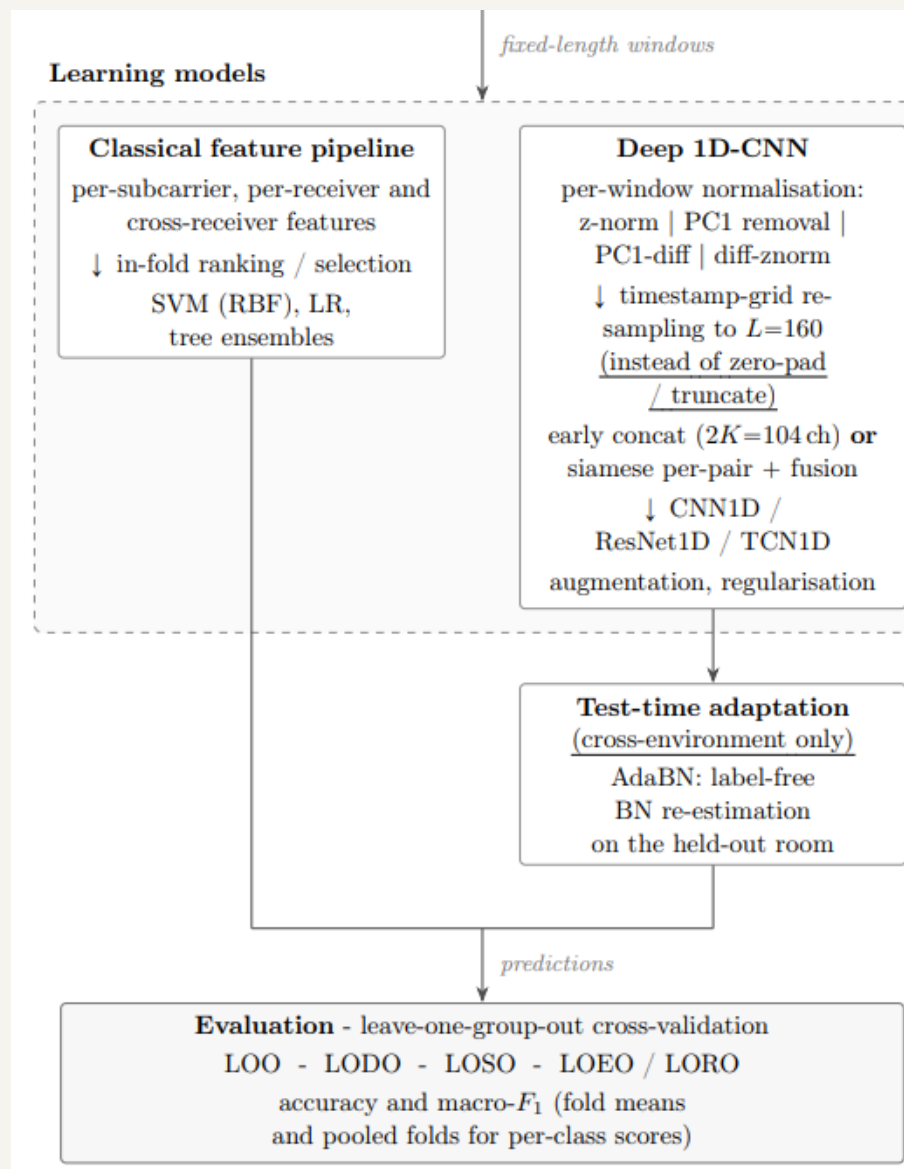
Median Filter example with $K=4$



Wavelet filter example







CLASSES

Class	Type	# Samples	Physical realisation
walk	continuous (≈ 10 s)	24–26 per environment	Walking around the room
prepare_meal	continuous (≈ 10 s)	24–26 per environment	Standing and performing meal-preparation arm movements
sit	transition (≈ 4 s)	70 per environment	Sitting down onto a chair
sit_up	transition (≈ 4 s)	70 per environment	Rising from the chair to standing
lying	transition (≈ 4 s)	70 per environment	Lying down on a mattress on the floor
lying_up	transition (≈ 4 s)	70 per environment	Getting up from lying to standing

Recording Days

Fold	Room	Recording days
A	mc208	15, 16 June, 1 July
B	sb207	18, 25 June
C	mf207	29, 30 June

Table 5.2: Preliminary double-pair experiment: cross-day macro- F_1 of the best model per family, six-class versus the merged three-class task (`sit+stand+lying+empty` $\rightarrow$ `no_action`). Means over the three held-out days; full sweeps and per-class breakdowns in Appendix A.3.

Family	6-class		3-class (merged)	
	Acc	F_1	Acc	F_1
Classical (features + LogReg)	0.558	0.558	0.911	0.911
Deep learning (best CNN)	0.463	0.451	0.956	0.956
Chance level	0.167		0.333	

→ The collapse during the continuous routines is clear for the fine grained activities

Evaluation regime	Acc	macro- F_1
Controlled single-activity clips (LOO, 198 clips)	0.909	0.909
Continuous real routines (350 windows, 6 sessions)	0.589	0.445
<u>Per-class F_1 on the real routines</u>		
Generic: <code>stand</code> / <code>walk</code>	0.92	/ 0.65
Fine-grained: <code>pour</code> / <code>prepare_cafe</code>	0.53	/ 0.44
Fine-grained: <code>grab_counter</code> / <code>grab_above</code>	0.13	/ 0.00

→ The initial grid search could surpass 0.58 macro-f1 score for the cross-room generalization

Family	Best configuration	Mean F_1	Acc.	Worst room
Deep (dual)	gain correction + denoising + band-pass 0.3–8 Hz diff-znorm $W=3.5$ s TCN1D	0.579	0.592	0.530 (sb207)
Features (dual)	gain correction per-env std. ℓ_1 top-300 RF	0.558	0.567	0.532 (sb207)
Deep (single, CB00)	gain correction + denoising + band-pass 0.5–10 Hz znorm $W=3.5$ s TCN1D	0.541	0.547	0.507 (sb207)
Features (single, CB00)	gain correction per-env std. ℓ_1 top-300 RF	0.539	0.548	0.534 (sb207)



HAND-CRAFTED FEATURE BANK – GENERAL CATEGORIES

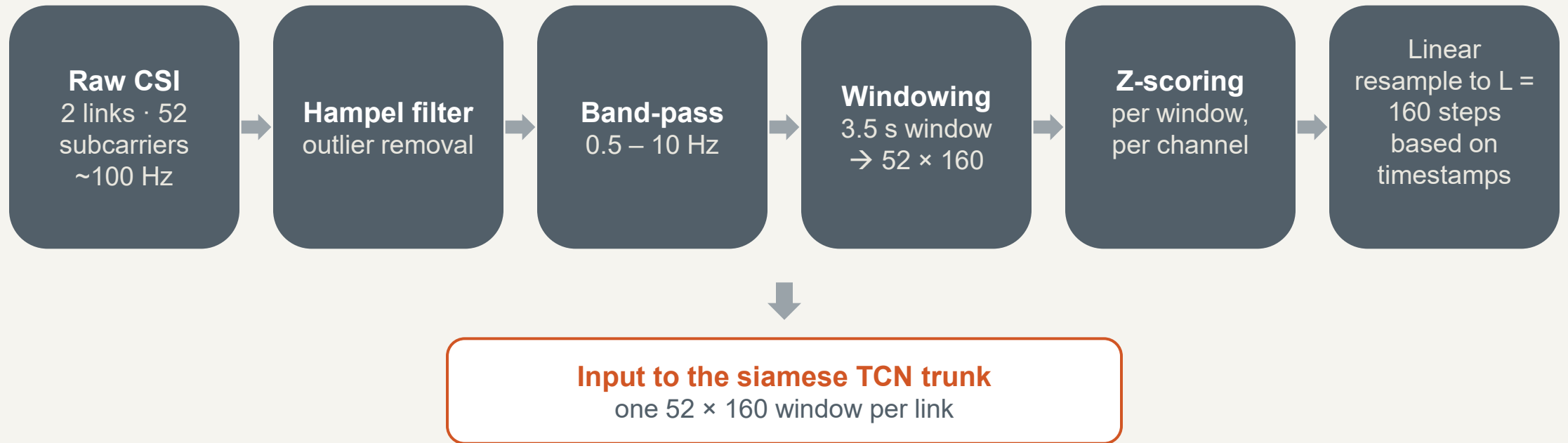
4,021 features per dual-pair window = (36 per-subcarrier x 52 + 8 RX-level) x 2 pairs + 53 cross-RX + 208 raw-energy

Category	Representative descriptors	Count
Amplitude statistics	mean, median, std, MAD, range, IQR, skewness, kurtosis	8 / subc.
Spectral shape	spectral entropy, dominant freq., >5 Hz ratio, total power, peakedness	5 / subc.
Band powers	0–1, 1–3, 3–8, 8+ Hz and 0.1–0.5, 0.5–2, 2–5, 5–10, 10–25 Hz	9 / subc.
Autocorrelation	lags 0.25–4 s; profile mean and max over lags 0.2–2 s	7 / subc.
Temporal / complexity	zero-crossing rate, Hjorth mobility & complexity, sub-window energy std, permutation entropy	5 / subc.
Breathing band	peak-to-noise ratio and dominant frequency in 0.1–0.5 Hz	2 / subc.
Receiver-level summaries	variance entropy, Gini, motion-event rate & duration, top-5 energy fraction, breathing ratio	8 / pair
Cross-receiver	per-subcarrier Pearson correlation between links, breathing-band coherence	53
Raw energy	per-subcarrier variance and mean-square of un-normalised amplitude	208

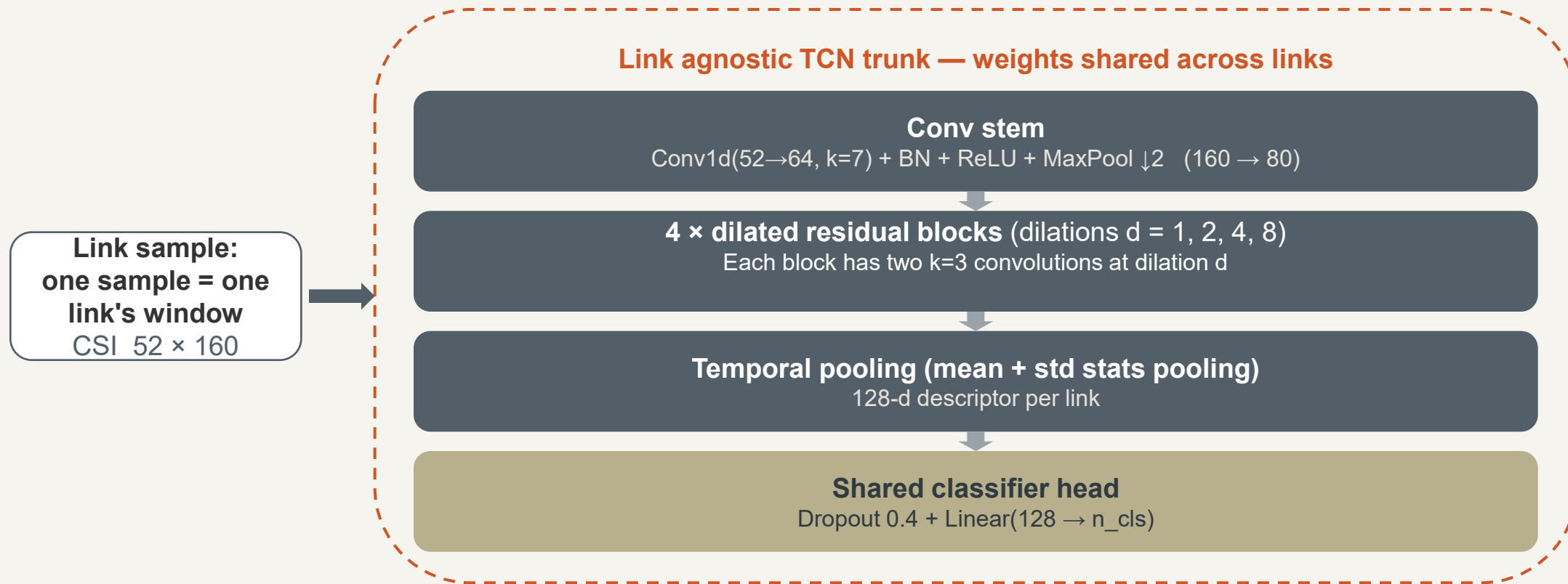
MAIN EXPERIMENT – FIRST SIAMESE TRUNK RESULTS

→ The experiment that led to using the Siamese shared trunk

Model	Mean F_1	Ensemble F_1
TCN, stacked 104-ch input (scratch)	0.532	0.574
Siamese shared trunk (scratch)	0.595	0.633
Siamese shared trunk + masked-recon pretraining	0.603	0.639
Pretrained transformer (MAE fine-tune)	0.485	0.495
Transformer (scratch)	0.303	0.316



FINAL MODEL – ARCHITECTURE - TRAINING



TRAINING RECIPE

Optimisation AdamW, lr $1e-3$, weight decay $1e-3$ - 30 epochs - batch 32

Schedule / loss cosine annealing; cross-entropy with label smoothing 0.05

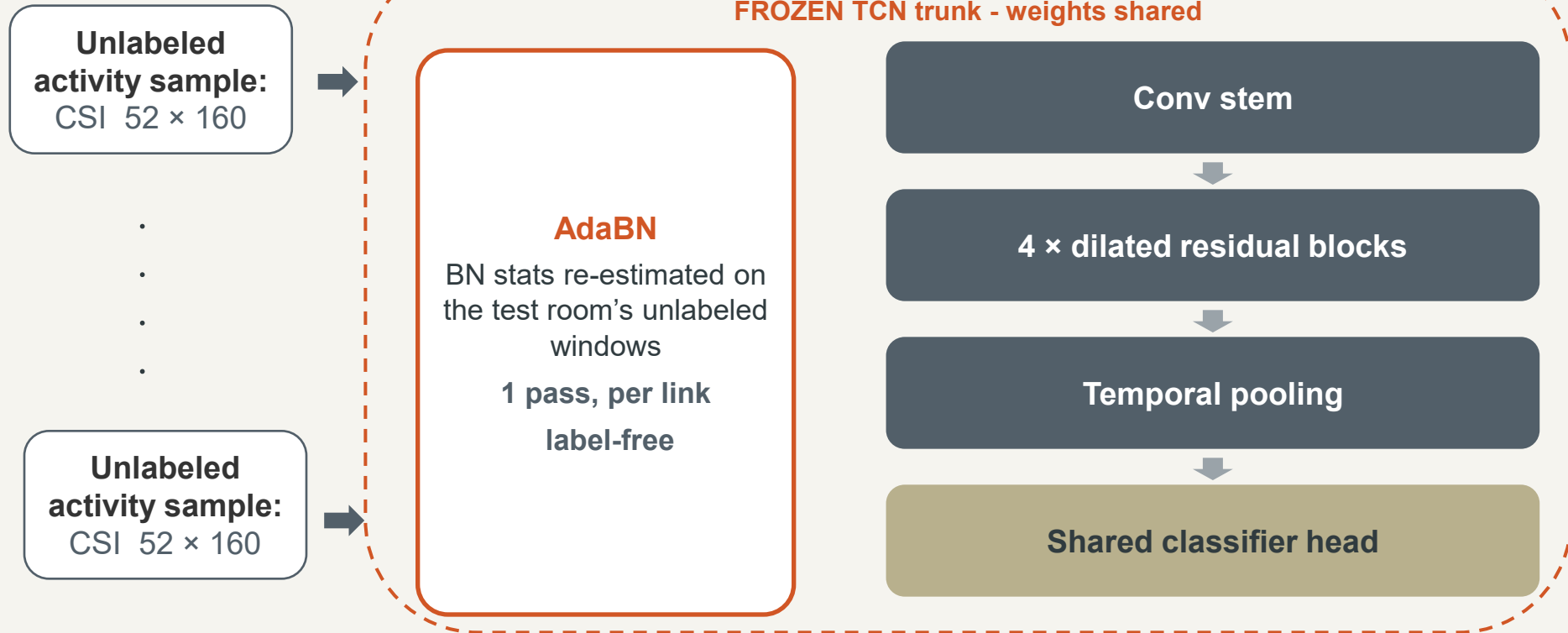
Masking subcarrier + time masking, $p = 0.7$ (3–8 subcarriers / spans of 8–20 steps)

Same-class mixup $p = 0.5$, $\lambda \in [0.6, 0.95]$ - partner may come from the other link: a deliberate cross-geometry regulariser

3-seed ensemble

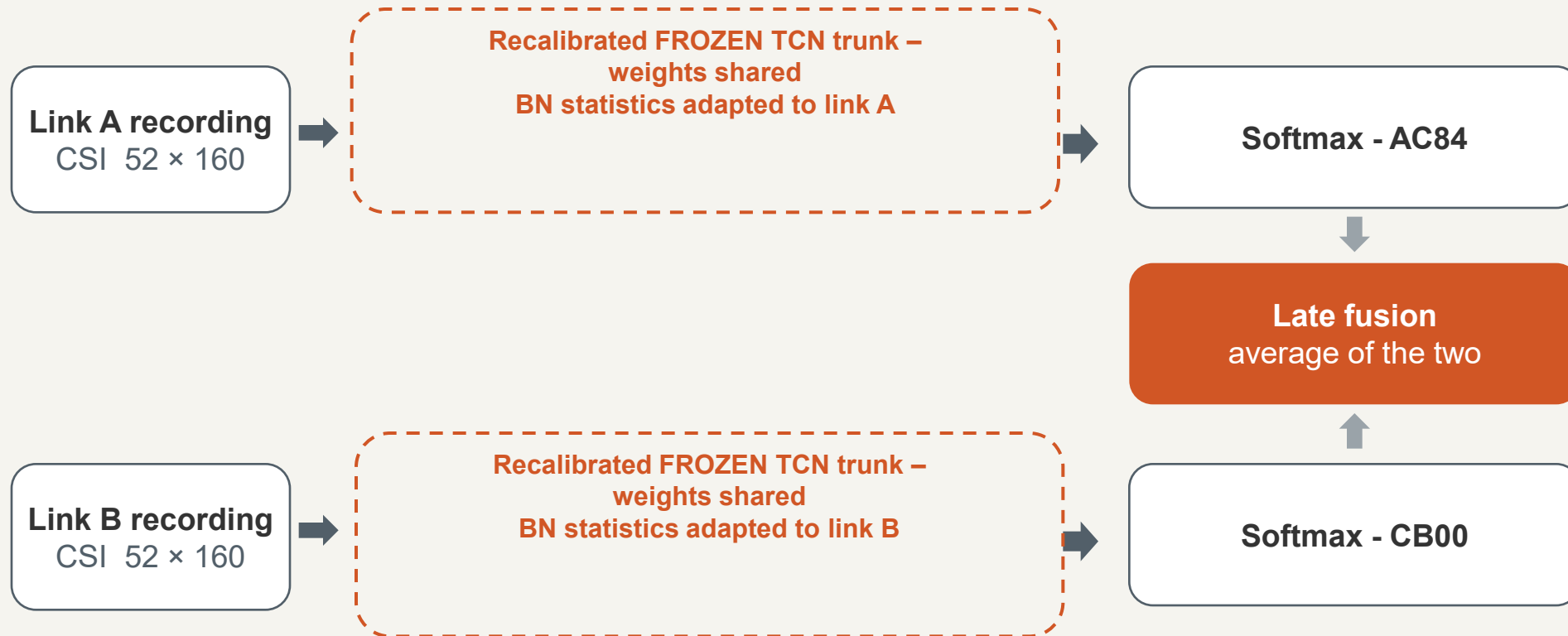
FINAL MODEL – UNLABELLED TEST ROOM ADAPTATION REAL TIME

One sample
per class:



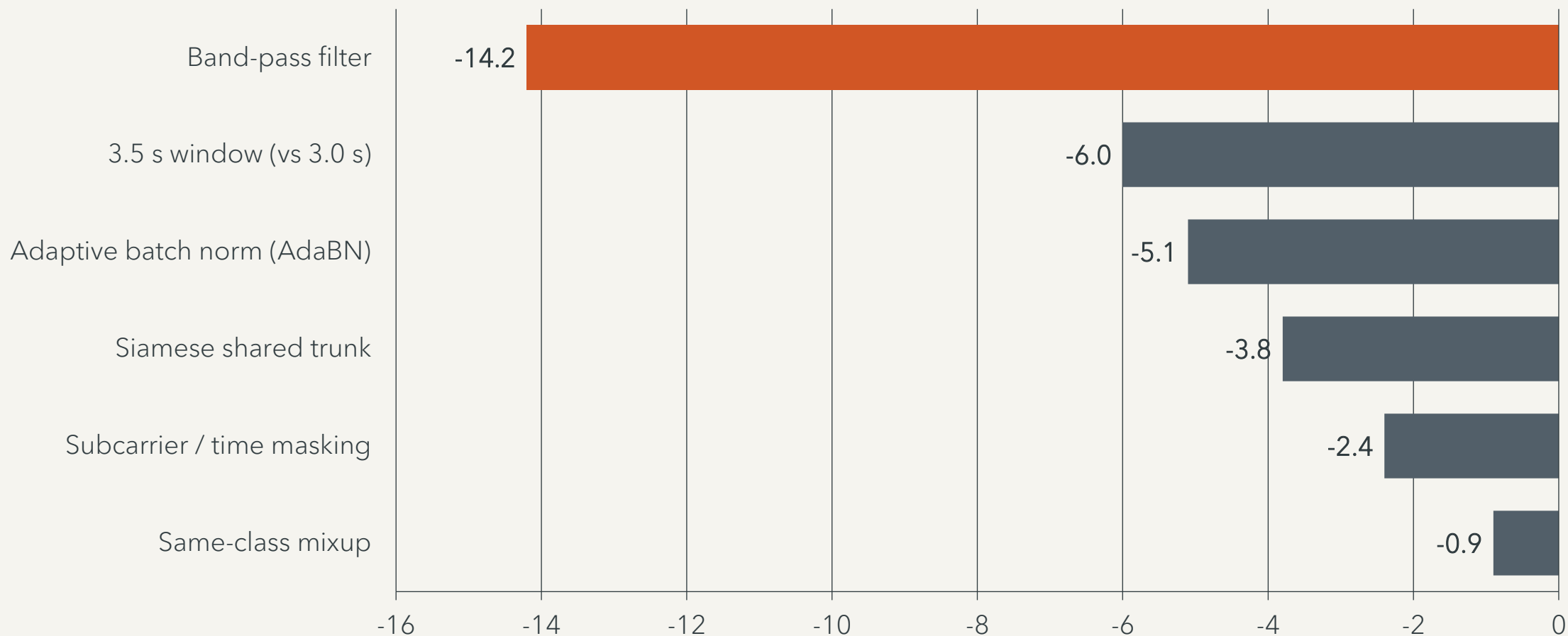
AdaBN happens per link so two links are no longer the same network: same weights but different BN statistics

FINAL MODEL – ARCHITECTURE INFERENCE



COMPONENT ABLATION – WHAT ACTUALLY MATTERS

Points of six-class macro-F1 lost when each component is removed



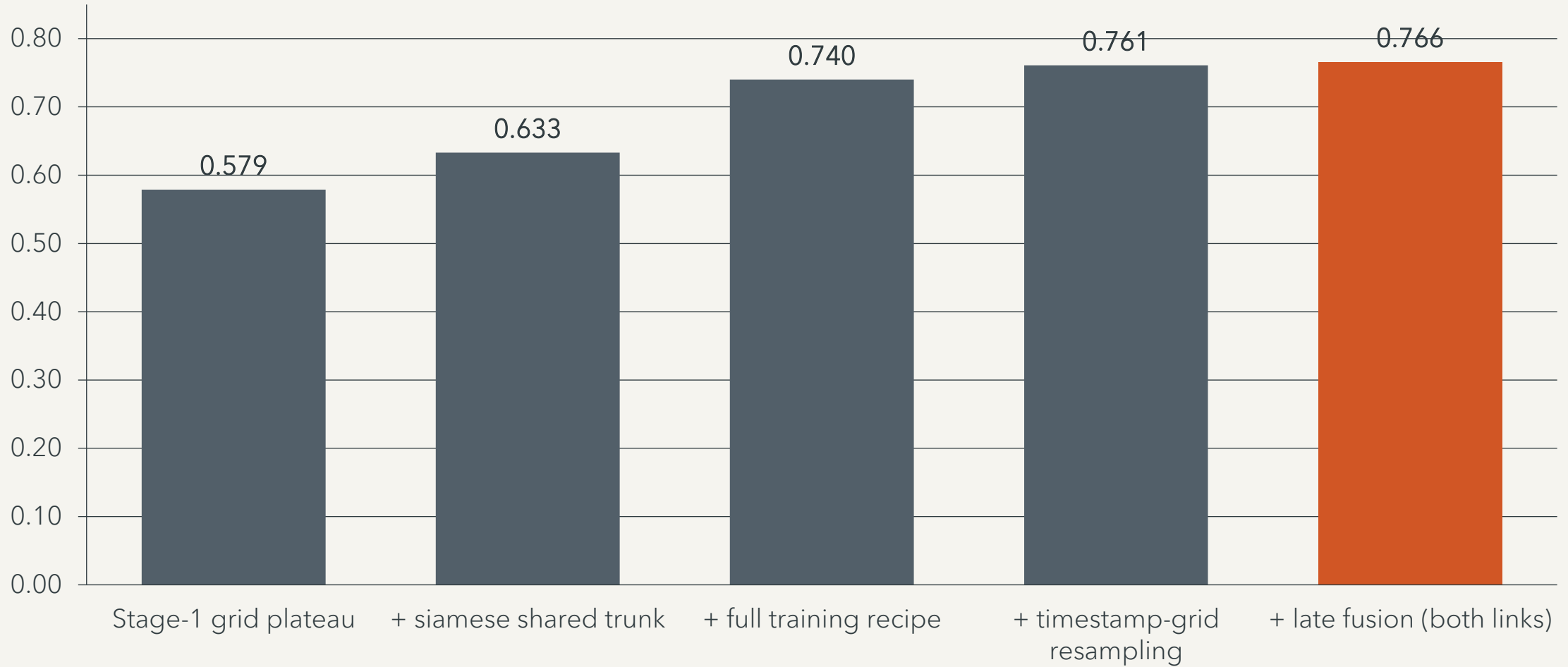
→ Using the timestamp resampling into $L=160$ is the clear winner

Sampling	6-class F_1	4-class F_1
Timestamp grid, $L=160$	0.761	0.882
FFT resample, $L=160$	0.749	0.879
Linear index grid, $L=160$	0.744	0.876
Timestamp grid, $L=128$	0.743	0.891
Timestamp grid, $L=96$	0.740	0.882
Linear index grid, $L=128$	0.735	0.887
Linear index grid, $L=96$	0.732	0.878
Band-pass after resampling, $L=160$	0.708	0.865
Linear index grid, $L=320$	0.696	0.857
No resampling (zero-pad)	0.663	0.858
No resampling (per-window band-pass)	0.637	0.858



WHAT UNLOCKED CROSS-ENVIRONMENT GENERALIZATION

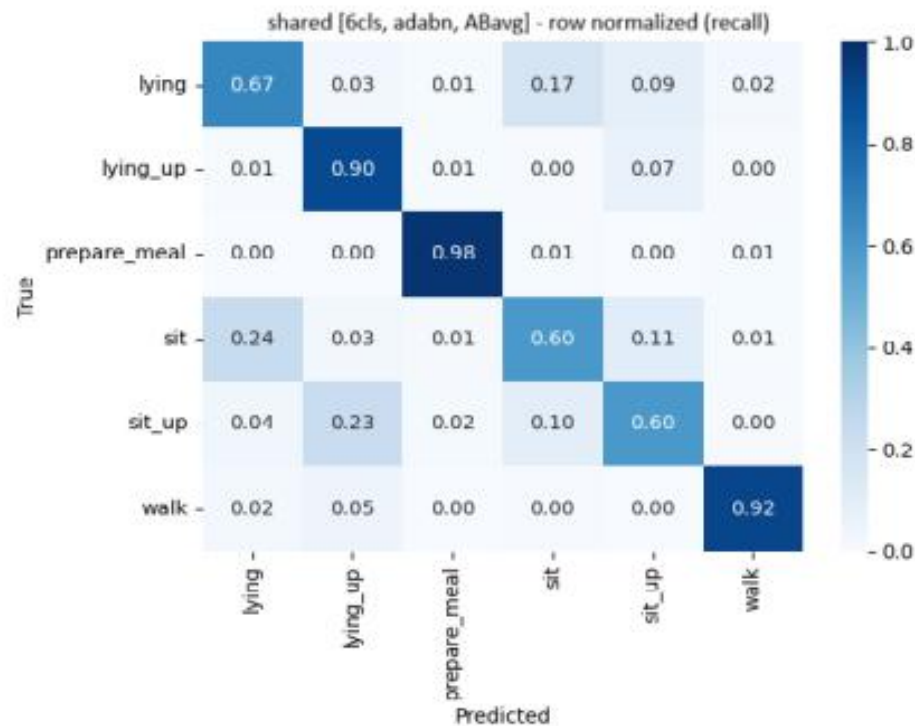
Leave-one-room-out macro-F1, six classes (mean over three held-out rooms)



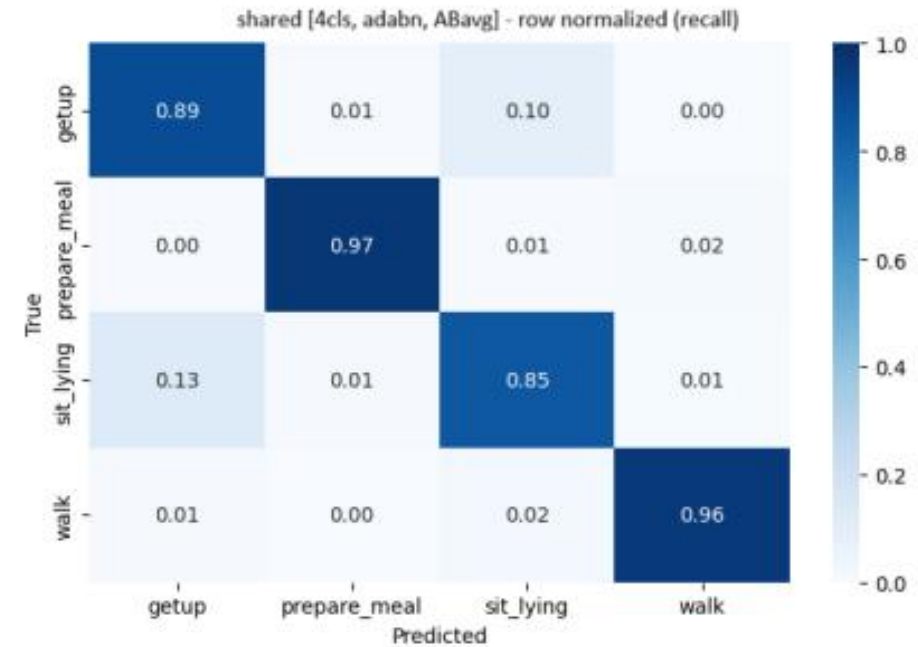
MAIN EXPERIMENT – CONFUSION MATRICES DOUBLE PAIR

→ The confusion between chair and floor is clear even for the double pair

6-CLASS



4-CLASS





MAIN EXPERIMENT – CLASSIFICATION REPORT DOUBLE PAIR

→ The classification reports for the double pair late fusion best model

6-CLASS

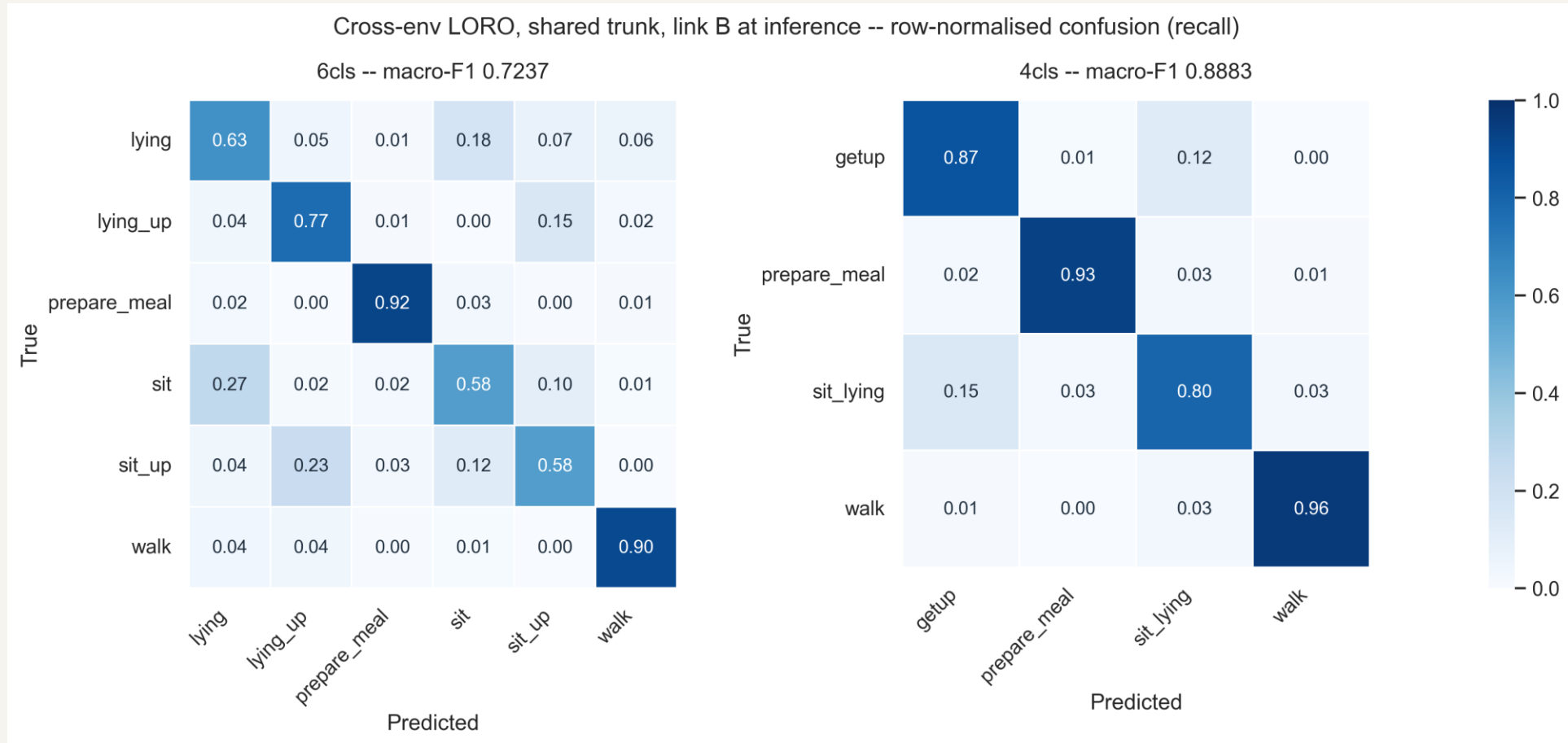
Activity	Precision	Recall	F1-Score	Support
Lying	0.68	0.67	0.67	210
Lying Up	0.72	0.90	0.80	210
Prepare Meal	0.95	0.98	0.96	210
Sit	0.67	0.60	0.63	210
Sit Up	0.69	0.60	0.64	210
Walk	0.95	0.92	0.93	210

4-CLASS

Activity	Precision	Recall	F1-Score	Support
Get Up	0.86	0.89	0.87	420
Prepare Meal	0.98	0.97	0.97	420
Sit/Lying	0.87	0.85	0.86	420
Walk	0.97	0.96	0.97	420

MAIN EXPERIMENT – CONFUSION MATRIX – LINK B INFERENCE

→ The confusion matrix for the suggested architecture of both links for training and link B for inference.



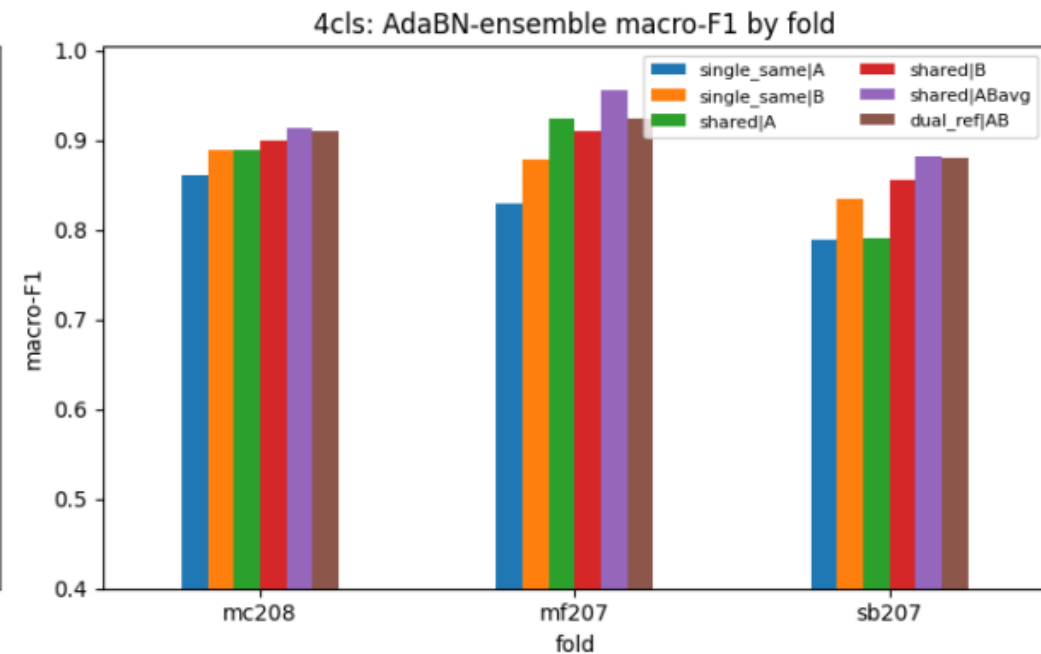
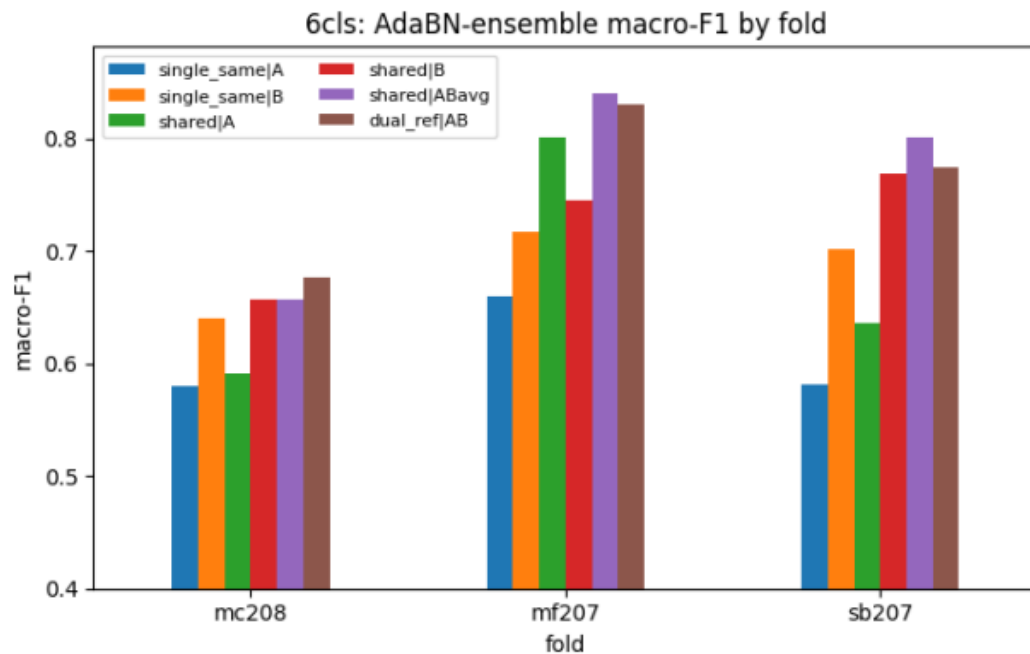


MAIN EXPERIMENT – PER ROOM RESULTS

→ Shared Abavg is the winner

6-CLASS

4-CLASS





MAIN EXPERIMENT – NO_ACTION EXTENSION

→ The no_action class successful extension

Training	Test-time input	Macro- F_1 (mean over folds)		no_action F_1 (pooled)
		7-class	5-class	7-class / 5-class
Shared trunk (both links)	Both links, prob. average	0.777	0.917	0.948 / 0.946
Shared trunk (both links)	Link B only	0.734	0.881	0.905 / 0.910
Single link B	Link B only	0.711	0.862	0.920 / 0.923
Shared trunk (both links)	Link A only	0.692	0.848	0.869 / 0.873
Single link A	Link A only	0.627	0.830	0.891 / 0.901



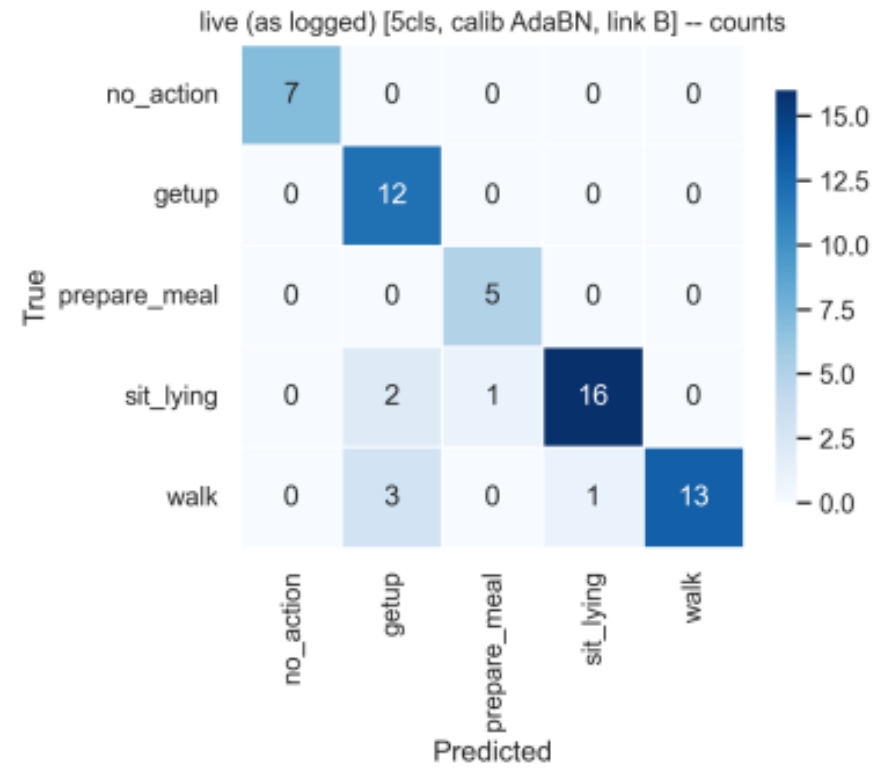
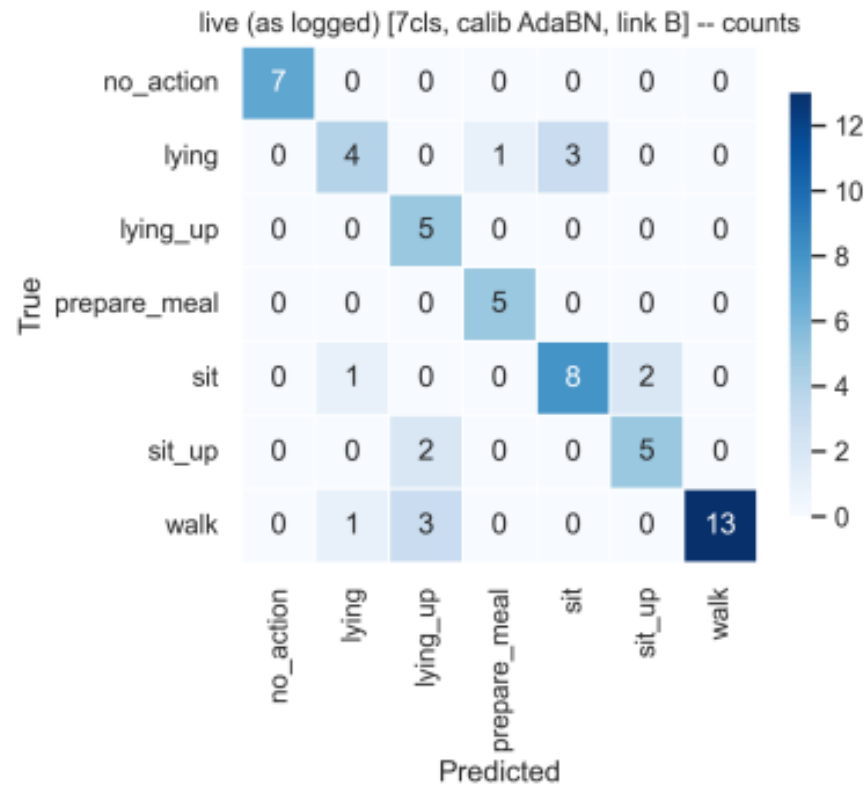
MAIN EXPERIMENT – LIVE SESSIONS

→ The model performs accordingly in the live cued sessions

Session	Deployment	Task	n	Acc	macro- F_1	LORO F_1
10 August, room e212	dual, prob. average	7-class	35	0.8	0.801	0.777
same windows, merged	dual, prob. average	5-class	35	0.943	0.948	0.917
23 July, living room	single, shared trunk with link B	7-class	60	0.783	0.779	0.734
same windows, merged	single, shared trunk with link B	5-class	60	0.883	0.899	0.881

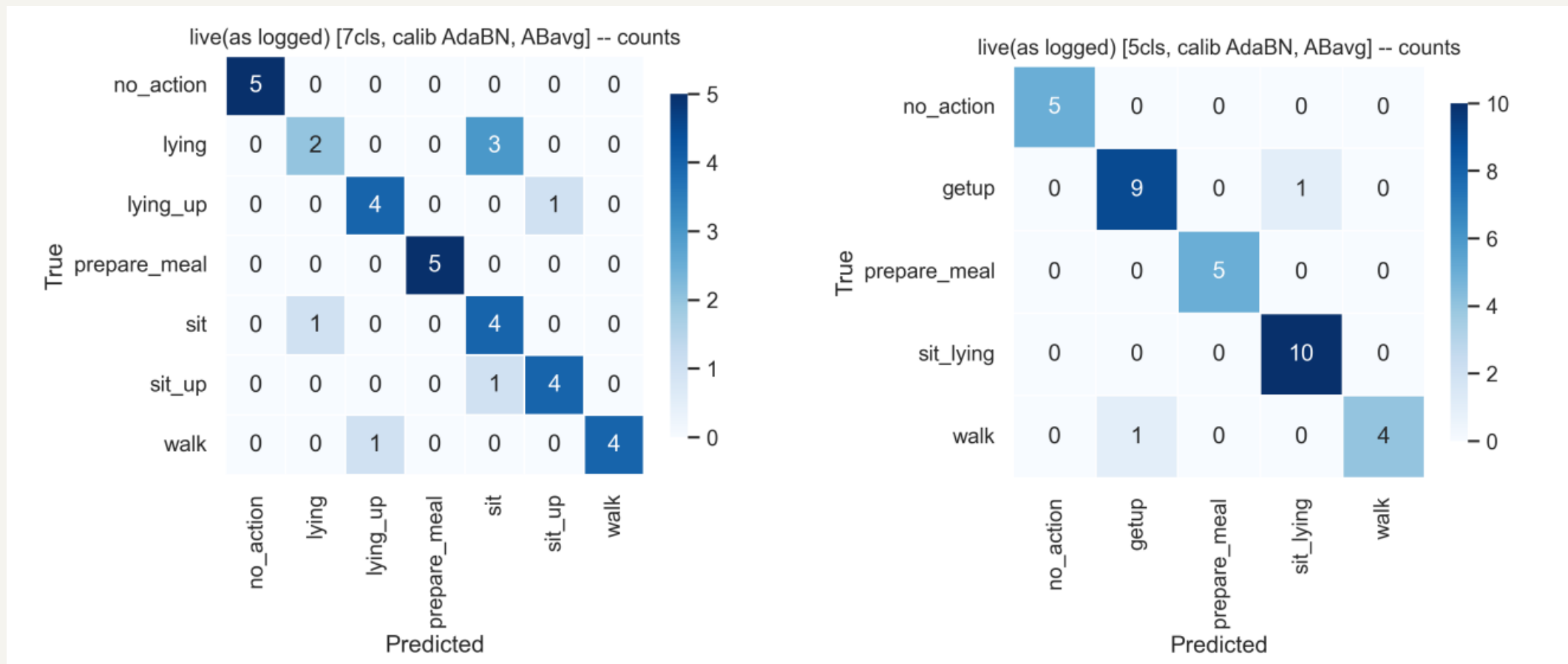
MAIN EXPERIMENT – LIVE SESSIONS CONFUSION

→ 23 July, single pair session confusion matrix for both granularities



MAIN EXPERIMENT – LIVE SESSIONS CONFUSION

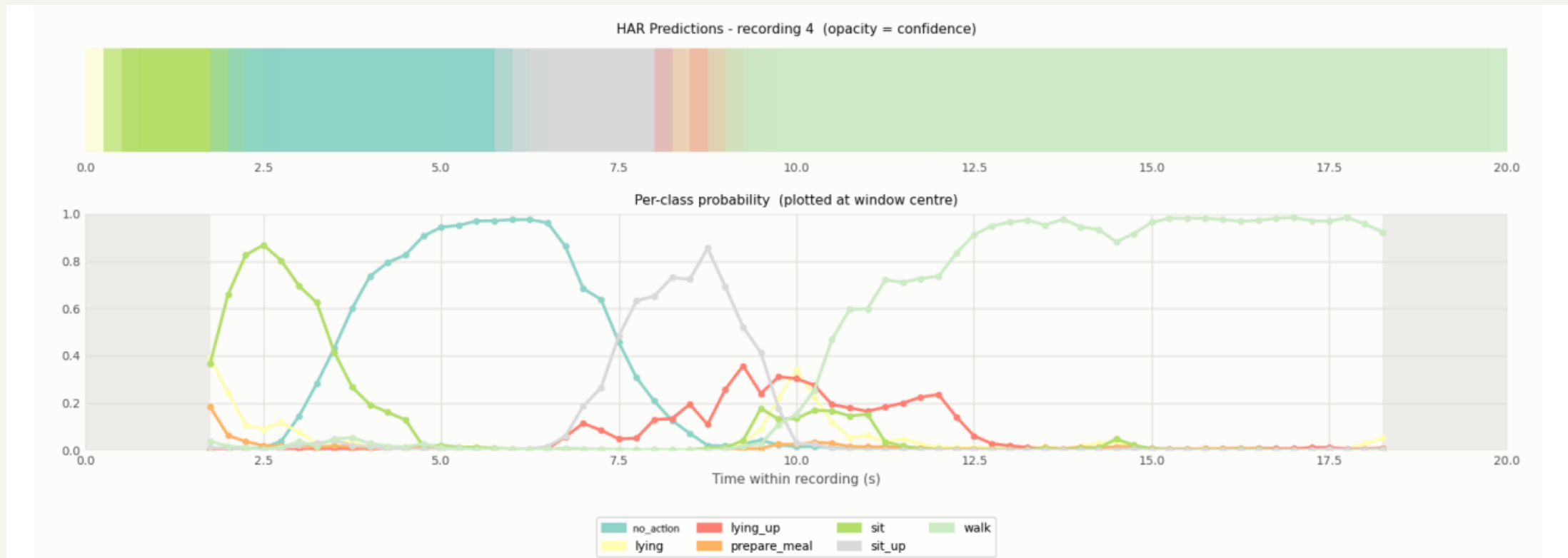
→ 10 August, double pair session confusion matrix for both granularities



MAIN EXPERIMENT – LIVE CONTINUOUS SESSIONS

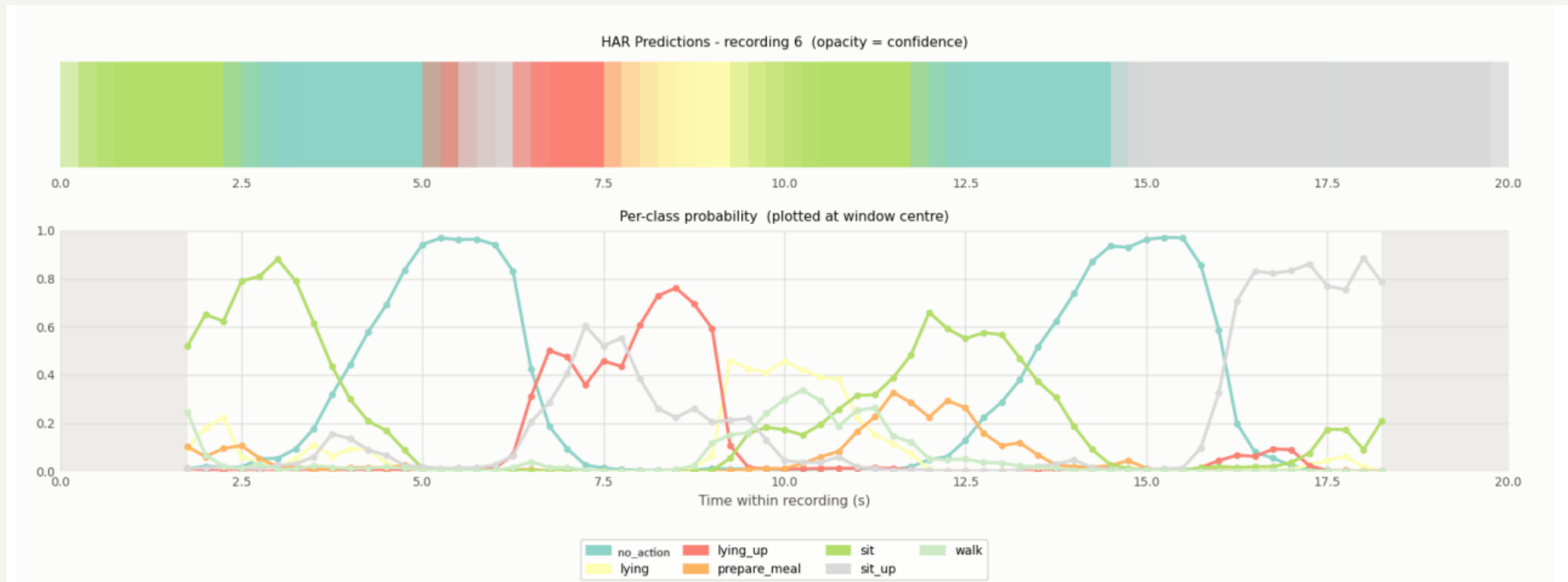
→ GT: lying → no_action → lying_up → walk

→ Lying is slightly confused with sitting in the beginning and lying_up is confused with sit_up after the idle state



MAIN EXPERIMENT – LIVE CONTINUOUS SESSIONS

- GT: lying → no_action → lying_up → sit → no_action → sit_up
- The stream still follows the sequence, except for lying which is misclassified as sit



MAIN EXPERIMENT – LIVE CONTINUOUS SESSIONS

→ GT: walk → prepare_meal → sit → sit_up → walk
 → All actions are identified, but some false actions are predicted between transitions;
 e.g., before prepare_meal, a lying and sit action is falsely predicted

